Ordinary Binary Polynomial Multiplication below $n \log n$

Controlled tensor routing on fixed tapes

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Abstract

We give a transfer from a fast equal-chunk tape interchange to ordinary dense multiplication in $\mathbb{F}_2[X]$. The main new interfaces are tensors of permutations on unequal even radices and a coupled cyclic-index to prime-residue index permutation. A flat perfect shuffle differs from its mixed-radix tensor by a low-endpoint cyclic rotation. A ragged parity-splitting network keeps the original even dimensions throughout its recursion. Controlled versions use completed target-only kernels and a temporary control gauge. A balanced CRT tree then needs only logarithmically many controlled tensor stages. The cyclotomic coefficient word can be enlarged to pay for polynomial-time address metadata without changing its small-word arithmetic regime. Together with the characteristic-two local Fourier compiler and the explicit interchange and endpoint-kernel constructions in the appendices, this gives the bound $O(n(\log n)^{1-2^{-38}})$ in the deterministic fixed-finite-tape bit model. This is a research preprint open for mathematical review; no external verification or priority is claimed.

AI assistance and formal verification. This work was developed with substantial generative AI assistance in mathematical exploration, algorithm and proof development, writing, coding, and literature review. OpenAI Codex [13] assisted release preparation and source inspection. Historical model versions were not consistently recorded; no complete model-level attribution is claimed. Complete Lean formalization is unfinished and planned for a later update. This manuscript is a research preprint open for mathematical review.

Status and scope. The input and output are ordinary monomial coefficient lists over $\mathbb{F}_2$. The function is ordinary polynomial multiplication. We count all address processing, padding, scalar preparation and record movement. The finite motif, interchange, endpoint-kernel and ordinary-I/O proofs are expanded in this manuscript and the selected proof-support records. The whole composition must be read as an actual tape construction. Finite checks below corroborate identities; they do not prove asymptotic running times. The companion manuscript, *Ordinary Polynomial Multiplication over Every Fixed Finite Field*, is provided alongside this paper in the review release.

Relationship to the companion papers. The accessible companion, *Exact Binary Polynomial Multiplication from Fast Pascal Transforms*, develops binary prefix/Taylor evaluation and a trace-pairing residue decoder. The present construction shares finite-network and equal-chunk interchange antecedents but uses a different outer route: a cyclotomic host, coefficient CRT components, a balanced CRT routing tree and ordinary cyclic-product reconstruction. It does not invoke the Pascal companion's complete multiplier as a subroutine. The general finite-field companion combines the binary bound proved here with its

odd-characteristic construction. The present proof does not depend on that combined consequence, so this relationship is not circular.

Prior work. An earlier fixed-prime benchmark is

$$O_p(n \log n \, 8^{\log^* n}),$$

due to Harvey, van der Hoeven and Lecerf [10]. Harvey and van der Hoeven supply the unconditional cyclotomic host and an $O(n \log n \, 4^{\log^* n})$ fixed-prime polynomial bound [1]. Their $O(n \log n)$ finite-field theorem treats the uniform-field problem under an explicit near-one Linnik hypothesis [2]. The present fixed-binary-field construction builds on their unconditional cyclotomic host and supplies ordinary coefficient reconstruction within its stated tape cost. Harvey's October 2026 discussion identifies fixed-field polynomial multiplication as a separate problem from the new integer claims [11, Section 4]. Van der Hoeven and Larrieu exploit Frobenius symmetry in finite-field Fourier transforms, with applications to binary polynomial multiplication [12]. Their extension arithmetic and conversion costs are part of that analysis. Our stated all-length bound requires the complete ordinary-input/output and routing accounting developed here. The 2026 interchange preprint supplies the finite-network tape method [5]; its global integer multiplication theorem is not needed. The exact-complex Fourier paper supplies the local Newton/Toeplitz and borrowed-coordinate mechanism [6]; its complex scalars and unit-cost indexing do not give a finite-field bit theorem directly. It also states an explicit exact complex-polynomial multiplication corollary in that arithmetic model. The corollary retains its specified-root and address-cost assumptions and expressly excludes finite fields and bit complexity. Our claimed coefficient field and machine model therefore require the separate transfer proved here. Cantor–Kaltofen [3] give classical formal convolution circuits over arbitrary coefficient algebras. We print an elementary characteristic-two triadic circuit with looser secondary logarithms to make our required generation bound explicit. Arbitrary-size permutation networks have substantial prior art, including [7]. We make no novelty claim for their abstract existence. Our required statement is a priced tensor execution on the existing mixed-radix tape, including controls and dummies. Current follow-up work also studies larger integer-multiplication exponents [8]; a separate repository states a finite-machine theorem for multiplication of natural numbers [9]. We do not use those claimed end-to-end bounds as premises. The cited primary and related sources are listed in the bibliography; this is not an exhaustive priority determination.

Together these works provide the host arithmetic, finite-network methods and transform mechanisms used in this paper. The specific advance claimed here is an ordinary binary polynomial multiplier with a fixed positive saving below the logarithmic exponent one, obtained by completing the controlled mixed-radix and CRT routing transfer. The preceding fixed-field bound and the conditional uniform-field theorem explain why that threshold is a meaningful target. The tiny saving is a theoretical asymptotic statement; the earlier algorithms and their structural contributions remain the foundations of this construction.

1 Model and supplied kernels

One machine has a fixed finite alphabet and a fixed finite number of one-dimensional tapes. Heads move by one cell per step. Shape parameters are ordinary binary integers. Arrays occur in lexicographic order. A record has D arbitrary bits, and V denotes the entire bit volume. Public addresses are positions of records, never payload controls.

We reconstruct the following interchange and endpoint kernels in the appendices, with further implementation detail in the companion dossier. Their dependency direction is explicit: the rational-frame interchange uses the binary scalar motif and ordinary scans; the

endpoint kernel then uses interchange and the finite ternary-phase word. The interchange proof never invokes H2 or a polynomial multiplier.

Proposition 1 (Interchange and native endpoint kernels)**.** *There is a fixed $\tau = 1 - 2^{-36} < 1$ such that equal address chunks in $[P]\times[2^u]\times[G]\times[2^u]\times[B]$ can be interchanged in $O(Vu^\tau)$ bit time, uniformly in positive P, G, B. Put $\gamma = 1/(2-\tau) < 1$; aggregate rectangular exchanges have cost $O(VL^\gamma)$ for total address width at most L.*

Let n_i be even native radices, $M = \prod_i n_i$, $L = \lceil \log_2(2M) \rceil$, and

$$L^3 \le n_i \le R, \qquad \log R = O((\log L)^c).$$

On each adjacent pair $(2a, 2a+1)$ put $H = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The tensor of these canonical native H maps has cost $O(MDL^\gamma \operatorname{polylog} L)$, uniformly for external $D \ge 1$. All private padding and work-tape returns are included.

Appendix D supplies interchange directly, re-instantiating the method of [5]; Appendices A and B prove the endpoint consequence. Thus the fast interchange statement is reconstructed here, rather than left as an unproved existence premise of the multiplication transfer.

The native kernel uses the quotient labels as carry guards, rather than individually padding the native radices. A frozen endpoint mask on u digits is a constant word of weighted integer rotations plus exact sparse boundary repair. A large untouched suffix supplies a dock and exponentially many opaque lanes. Its numerical padding survives the complete binary kernel, including every internal $\mathbb{F}_{16}$ letter. The supplier proof must retain those lifetimes.

Deriving the aggregate rectangle bound. For completeness, interchange two adjacent binary fields of widths a, b, with $a + b \le L$. Choose an integer threshold $u = \lceil L^{1/(2-\tau)} \rceil$. While both residual fields have width at least u, use the block-exchange Euclidean algorithm. If $a \le b$, split the second field into its first a bits and the remaining $b - a$, exchange the two equal a-bit fields, and retain the already placed prefix. The other case is symmetric. Each step removes its shorter width v from the unresolved total. Thus the sum of those v is at most L, and, since $\tau - 1 < 0$,

$$\sum_{v \ge u} O(Vv^\tau) \le O(Vu^{\tau-1}) \sum v = O(VLu^{\tau-1}).$$

When one residual field has width below u, move its bits individually through the other field, at $O(V)$ per bit, using streamed two-row chunk interleaves. Each suffix block is copied as a block, with amortized counters. The last cost is $O(Vu)$. Move a right-hand short field's bits first to last to the front of the long field; move a left-hand short field's bits last to first after it, retaining already placed bits. Both routes preserve the short field's internal order. Both terms are $O(VL^\gamma)$ for $\gamma = 1/(2-\tau)$. All calls keep the same full volume V and already placed fields are spectators. For rational fixed τ, an integer power comparison computes the threshold; its polynomial-in-L setup fits in the array volume. To exchange native aggregate ranges, pad each whole range once to a binary range (combined factor below four), perform this exchange and crop the nested valid rectangle after completion. There is no separate pad for each native axis.

2 A tensor perfect shuffle

Write $n_i = 2k_i$ with the first digit most significant. Define

$$S_i(k_i h_i + \ell_i) = 2\ell_i + h_i, \quad h_i \in \{0, 1\}, \quad 0 \le \ell_i < k_i.$$

Also define $d_M(X) = 2X \bmod (M-1)$ for $X < M-1$ and fix $M-1$. Splitting the flat stream into its two halves and interleaving records implements d_M in $O(V)$ time.

Lemma 2 (Post-shuffle endpoint correction)**.** *The tensor $\bigotimes_i S_i$ and its inverse have cost $O(VL^\gamma \operatorname{polylog} L)$ under Proposition 1.*

Proof. For $x_i = k_i h_i + \ell_i$ and incoming carry $c \in \{0,1\}$,

$$2x_i + c = n_i h_i + (2\ell_i + c).$$

The outgoing carry is exactly h_i, independent of c. End-around doubling therefore has output $y_i = 2\ell_i + h_{i+1}$, cyclically in i. Consequently

$$\bigotimes_i S_i = r \circ d_M, \qquad r(y)_i = 2\lfloor y_i/2 \rfloor + (y_{i-1} \bmod 2).$$

This rotates low endpoint bits *after* the flat shuffle; it does not require a cheap rotation of the original high-half bits.

Index the axes from zero. Reverse endpoint bits on $[0, t-1)$ and then on $[0, t)$. A bit at $i < t-1$ moves first to $t-2-i$ and then to $i+1$; the last bit moves to zero. This is precisely r. For a reversal $[a, b)$, a left-half interval $I = [c, d)$ pairs with $J = [a+b-d, a+b-c)$, in reverse local order. Partition the left half into such intervals of length $u \le h$, where

$$\ell_R = \lceil \log_2 R \rceil, \qquad h = \min\{\lfloor t/16 \rfloor, \lfloor L/(40\ell_R) \rfloor\}.$$

The complement of two selected intervals has at most three linear intervals. Since $t \ge 16h$, one has at least u axes; choose a contiguous u-axis tile auxiliary W there.

Place the two selected banks X, Y and W at the front in that order. Extract them successively from the remaining axis list: at each step exchange the whole preceding aggregate gap with the requested interval. At most three adjacent aggregate rectangle exchanges suffice. All unselected axes keep their relative order; reverse the same exchanges after the batch. Each exchange crops its two aggregate pads before the next one and costs $O(VL^\gamma)$. No endpoint gather is performed.

With $\pi(j) = u - 1 - j$, the chronological endpoint masks $Y_{\pi(j)} \oplus= X_j$, $X_j \oplus= Y_{\pi(j)}$, $Y_{\pi(j)} \oplus= X_j$ exchange the paired bits. Each reads only the opposite frozen bank and uses W as its separate dirty tile auxiliary; its weighted offset uses the target bank's own native place values. Every quotient label is fixed. The three banks use at most $3L/40$ address bits. After padding the one remaining suffix to 2^s and borrowing a dock of width $c \le L/40 + 1$, its low index supplies $R_{\mathrm{op}} = 2^{s-c} D \ge 2^{9L/10 - O(1)} D$ opaque bits per record. This pays the weighted-mask metadata and exact boundary repair even when $D = 1$. There are no phase-role bits in these direct masks. The completed masks restore the dock and fix the low suffix index, so the suffix pad is cropped before the inverse layout.

Each mask costs $O(V(u \log R)^\tau + V)$. Since $\gamma > \tau$ and $h \log R \le L/40$, a batch costs $O(VL^\gamma)$. From $L - 2 < \log_2 M \le t \log_2 R$, $h \ge L/(80\ell_R)$ for sufficiently large admissible L. Together with $t \le L/(3 \log_2 L)$ this gives $O(\log R)$ batches. Summing proves the bound. Reversing the actual word gives the inverse; the bounded remaining range uses a direct finite branch. □

3 Permutations on exact even dimensions

An adjacent pair can select either I or the exchange matrix J. Over $\mathbb{F}_4$, both have a common word of exactly eight H letters interleaved with unit diagonals. Tensoring a slot uses the supplied canonical kernel. The diagonal coefficient factors by local axes; its lexicographic

list is generated by prefix products in $O(M)$ fixed-field operations. The completed word is binary. Its lane pad can be removed only at that completed boundary.

Odd-start adjacent matchings are conjugates by coordinatewise $+1$. Explicitly, if $T(x) = x+1 \bmod n$, an odd-start pair $(i, i+1)$ is TKT^{-1} with K switching $(i-1, i)$. Set $K = I$ on every irrelevant pair, in particular $(n-2, n-1)$ whose conjugate would wrap to $(n-1, 0)$. Thus the completed matching has no cross-boundary or wrap switch, even when its temporary K crosses a ragged parent boundary. At a synchronized odd-start stage, translate every axis in the fixed target set, including axes whose matching is identity. Choose $K = I$ on those axes; conjugation then cancels their translations. Thus the translated set, flat offset and source mask are independent of switch settings, including any control address in the controlled compiler. This selective translation costs $O(V)$: flat rotation by the sum of the selected native place values agrees except when a selected digit wraps. The exceptional fraction is at most $t/L^3 \le L^{-2}$. Rotate the good-source mask with the payload. For bad output slots, compute $x = P^{-1}(y)$ and the correct destination $T(x)$, sort those records by destination, and refill the holes. Since $P = T$ on good sources and both are permutations, $P(B) = T(B)$. This does not assume that the bad source set is invariant. Keys, record sorting and schoolbook decoding cost

$$O(\delta ML(D+L) + \delta ML^2) = O(V).$$

Here the L^2 exception-address bound does not perform t full L^2 divisions. If $b_i = \lceil \log_2 n_i \rceil$, then $\sum_i b_i \le 2L$. Streaming a current at-most-L-bit numerator through a b_i-bit remainder register divides by n_i in $O(Lb_i + L + b_i)$ tape steps. Summing gives $O(L^2)$ for all digits; the weighted Horner reconstruction has the same bound.

The source-good mask also has a linear implementation. Starting with one bit, append native factors in their physical order. For each bit a of the previous mask, print a for the $n_i - 1$ nonmaximal values and zero at the maximal value if axis i is selected; otherwise print n_i copies of a. Counter reset work is at most its ensuing n_i-bit output block. Since the intermediate mask lengths grow by factors at least two, their sum is below $2M$. Rotate that mask with the flat data permutation, so it marks exactly the output holes $P(B)$. Compute and copy a full key only at a bad record. A monotone output counter otherwise advances amortized in $O(M)$ total time. Stable binary radix sorting in L passes costs $O(KL(D+L))$ for $K \le \delta M$ bad records, including the full payload and header scans. In each pass stage one fixed-width item, remember its selected key bit, and copy the item to the corresponding stable zero/one bucket. Concatenate the buckets and rewind; a constant number of tapes and $O(K(D+L))$ steps suffice per pass, including staging and returns. Binary counters have amortized linear cost over each scan; their per-item resets cost at most that item's width. Since the sorted destinations are exactly $P(B) = T(B)$, refill bad holes in that order; there is no L-bit key comparison at each good position. This establishes the stated bound even for $D = 1$. The negative translation uses nonzero-digit source masks and the same proof.

Lemma 3 (Ragged global-shuffle network)**.** *Every permutation of an original even n has a word of $O(\log n)$ levels consisting of adjacent switches and the* same *full S_n and S_n^{-1}. No recursive odd-sized child is padded.*

Proof. At a level, the residual permutation preserves *each* interval setwise in a public ordered partition into contiguous parents. Pair positions from each parent's start, leaving its last position unmatched if its size is odd. Use the same output pairing. The bipartite multigraph has input pair/singleton vertices, output pair/singleton vertices, and one edge per packet. Even parents give even cycles. An odd parent gives one path between its two singleton vertices, plus even cycles. That path has an odd number of edges. Its endpoint edge colors can both be forced to the global parity of the parent's last position.

Alternating edge colors give input switches E_{in} and output switches E_{out} so that the residual has identity

$$\pi = E_{\text{out}} S_n C S_n^{-1} E_{\text{in}}.$$

The full inverse shuffle stably splits global even and odd positions. For original even n, write this split as

$$p_n(x) = S_n^{-1}(x) = \begin{cases} x/2, & x \text{ even}, \\ n/2 + (x-1)/2, & x \text{ odd}. \end{cases}$$

A parent interval $[a, b)$ therefore has the two image fragments

$$[\lceil a/2 \rceil , \lceil b/2 \rceil) , \qquad [n/2 + \lfloor a/2 \rfloor , n/2 + \lfloor b/2 \rfloor) .$$

The even fragments concatenate to $[0, n/2)$ and the odd fragments to $[n/2, n)$, skipping empty fragments. The new contiguous block list is

[all parents' even fragments][all parents' odd fragments].

The endpoints depend only on n and the parent partition, independently of π and its switch colors. The residual C preserves each new fragment setwise. The recursive call is on this single n-wire residual with the whole updated partition. Thus each level adds only a fixed number of slot types to one recursive word. Track this order; do not gather siblings. Each fragment has size at most the ceiling of half its parent. After $\lceil \log_2 n \rceil$ levels all blocks are singletons. Each adjacent switch stage splits into its even-start and odd-start families. Short axes can continue passive singleton levels; their remaining shuffles cancel on unwind. □

Synchronize the networks for all axes to $\lceil \log_2 R \rceil$ levels. Lemmas 2–3 and the adjacent-switch words give a generic local-permutation tensor cost

$$O(VL^{\gamma} \operatorname{polylog} L). \tag{1}$$

This retains the parameter of the supplied kernels: each adjacent-switch word uses a fixed number of native-H calls, each shuffle has the same γ tariff, and there are $O(\log R)$ levels. Translations and fixed-field diagonal scans cost $O(V)$. Thus synchronization adds only a fixed power of $\log L$. Public local tables and network settings have polynomial-time sequential generation in R, L.

4 Controlled tensors and the control gauge

Partition axes into targets T and controls C. For each control address c let $\pi_i(c)$ be a permutation on target axis i. Controls are address coordinates, not input-dependent payload values.

Lemma 4 (Controlled compiler)**.** *Let $G(R, L) \geq 1$ bound the time to generate one outer $\mathbb{F}_4$ scalar coefficient at a current record address, including address decoding, control-gauge inversion, all required local tables and ragged settings. Assume this is a fixed polynomial-time sequential procedure with fixed tapes. Then the controlled map*

$$(c, x_T) \longmapsto (c, (\pi_i(c)(x_i))_{i \in T})$$

has bit cost

$$O((MDL^{\gamma} + MG(R, L)) \operatorname{polylog} L)$$

with fixed tapes and ordinary order. In particular $G(R, L) = O(D)$ gives (1)*. For the explicit CRT families below, $G(R, L) = O((R + L + 2)^{12})$; thus $D \geq (R + L + 2)^{64}$ suffices.*

Proof. First construct the completed subset kernel $K_T = H_T \otimes I_C$. Use a static common eight-H word for H versus I, tensoring each inner letter across all axes. The control axes may move *inside* this implementation. At its completed boundary, K_T fixes them. Insert controlled scalar stages only between completed K_T calls. The outer eight-H switch word is then valid on each fixed-control fiber. Its coefficient is $\kappa(x, c) = \prod_{i \in T} \kappa_i(x_i, c) \in \mathbb{F}_4^*$. We do not assume factorization by control axes.

The fragment-endpoint formulas in Lemma 3 make the partition and shuffle topology independent of the control address. Only the switch settings depend on c, so a single fixed slot schedule applies to all fibers. Use the full tensor shuffle on *all* axes. Give C identity switches. At an outer slot their temporary physical value is $q_e(c) = S_C^e(c)$, where e is the signed number of completed forward minus inverse shuffles. At each scalar stage enumerate the current native lexicographic addresses and decode logical $c = q_e^{-1}(x_C)$. Generate the target settings for this logical c. At every odd-start stage use the same translation of all axes in the fixed set T, with identity $K_i(c)$ on idle axes or fibers. The flat approximation can temporarily carry into a control coordinate on bad records. Complete its exact repair before reading any control-dependent setting. Completed subset kernels and translations then preserve the physical controls; a shuffle increments e in its prescribed direction. Induction over the word proves the fiber invariant. Descent and unwind balance, so $e = 0$ at exit and the controls are restored.

For the CRT affine families, one concrete generator decodes addresses by schoolbook division, computes the control CRT sums, generates the at-most-R local table, and constructs the pair graph by sequential scans. All axes, levels, records, and full-list lookups are bounded in the companion ledger by $O((R + L + 2)^{12})$ time. Fix the dominating polynomial $P_1 = (R + L + 2)^{64}$ before choosing D. For each record park its D bits locally, generate one coefficient, apply its fixed two-bit map, and advance. In general this scalar stage costs $O(M(D+G))$; visited work intervals and returns are included in the generator bound. There are only $\operatorname{polylog} L$ outer stages, proving the displayed general bound. The CRT generator just described has $G = O((R + L + 2)^{12})$, which the chosen word width dominates, so its scalar stage costs $O(V)$. No whole-array scan is nested inside a per-record lookup. This also aligns coefficients to the actual post-shuffle stream order. □

5 Enlarging the cyclotomic word

Lemma 5 (Uniform host initialization)**.** *From the binary input length n, for all sufficiently large n a deterministic fixed-tape procedure constructs a 2-admissible tuple $(q_0, \dots, q_e)$, its product $2n < N < 4n$, the attached common-multiple data, and $\alpha, \Phi_\alpha, \rho$ with $q_0 \mid \alpha$ and ρ a principal (N/q_0)-root. Its bit time is $N^{o(1)}$, including writing and verifying the factor and root data.*

Proof. Apply [1, Proposition 3.4] at $p = 2$ and target $2n$; its tuple and auxiliary data are part of the constructed admissible length. Apply Proposition 3.8 of that source to this tuple. Their respective time bounds are $2^{O((\log\log n)^2)}$ and $2^{O((\log\log N)^4)}$ at fixed $p = 2$. Both are $N^{o(1)}$ in the source's deterministic fixed-tape model. Write $L_0 = \lceil \log_2 N \rceil$ and $\ell = \lceil \log_2 L_0 \rceil$; its divisor has $\log \alpha = O(\ell^4)$.

Verification uses only polynomial-size setup operations. Trial division checks the emitted prime list and its product, since each prime is at most $R_0 = 2^{\ell^2}$; monic polynomial arithmetic checks the generated cyclotomic data. Put $K = N/q_0$. Check $\rho^K = 1$ and

$$\gcd(\rho^{K/q_i} - 1, \Phi_\alpha) = 1 \qquad (1 \le i \le e).$$

The squarefree quotient is a product of fields. These tests force order K in every component, hence principality by the geometric-sum identity. Modular powers, gcds and all certificates

cost a polynomial in α, R_0, L_0, which is $2^{O(\ell^4)} = N^{o(1)}$. Schoolbook arithmetic suffices in this setup; it uses no call to the proposed multiplier. □

If n is implicit in end-marked coefficient lists, counting and copying them costs $O(n)$ by amortized counters. The bounded initial length range uses the direct branch; no factorization is supplied as advice.

Use the unconditional host of [1]. Put $L_0 = \lceil \log_2 N \rceil$, $\ell = \lceil \log_2 L_0 \rceil$ and $R_0 = 2^{\ell^2}$. Its squarefree N has prime factors in $(L_0^3, R_0]$. The constructed divisor α uses $O(\ell^2)$ factors and satisfies

$$\log \alpha = O(\ell^4), \qquad -\log \frac{\varphi(\alpha)}{\alpha} = O(\ell^2 / L_0^3).$$

The distinguished prime q_0 divides α. Proposition 3.8 constructs ρ, a principal (N/q_0)-root in $\mathbb{F}_2[Y]/\Phi_\alpha(Y)$, as well as this cyclotomic polynomial. Then $\omega = \rho^{\alpha/q_0}$ is a principal $m = N/\alpha$ root. Indeed for $s = \alpha/q_0$ and $0 < j < m$, the old $K = sm$ character sum at exponent sj is s copies of the m-sum. Odd s is one in characteristic two, so that sum vanishes.

Append unused prime factors until $\alpha' = \alpha J$ satisfies

$$D' = \varphi(\alpha') \geq (R_0 + L_0 + 5)^{64}.$$

Only $O(\ell)$ factors are needed because each multiplies D by more than L_0^2. There are $\Omega(L_0/\ell^2)$ available factors. The first-crossing choice implies

$$\log \alpha' = O(\ell^4), \qquad \varphi(\alpha')/\alpha' > 1 - 1/L_0, \quad m'D' = \Theta(N), \quad m' = N/\alpha'.$$

We do not claim the enlarged divisor meets the original definition's literal numerical upper cutoff; these displayed estimates are sufficient. We retain the stronger density estimate as well. The original divisor has $O(\ell^2)$ prime factors, and enlargement adds only $O(\ell)$, all greater than L_0^3. Thus

$$\theta := 1 - \frac{D'}{\alpha'} = 1 - \prod_{p \mid \alpha'} (1 - p^{-1}) \leq \sum_{p \mid \alpha'} p^{-1} = O(\ell^2 / L_0^3). \tag{2}$$

This follows from the actual divisor construction in [1, Proposition 3.8, proof], rather than just its weaker admissible-divisor definition.

The substitution $Y \mapsto U^J$ defines a unital homomorphism

$$\mathbb{F}_2[Y]/\Phi_\alpha(Y) \longrightarrow S = \mathbb{F}_2[U]/\Phi_{\alpha'}(U),$$

because $\Phi_{\alpha'}(U)$ divides $\Phi_\alpha(U^J)$. Write ι for this homomorphism. It preserves principal-root character sums. The required root is $\omega' = \iota(\omega)^J$: it satisfies $(\omega')^{m'} = 1$. For $0 < j < m'$, split the old $m = Jm'$ terms into J equal periods to obtain

$$0 = \sum_{a=0}^{m-1} \iota(\omega)^{Jja} = J \sum_{b=0}^{m'-1} (\omega')^{jb}.$$

Odd J is one in characteristic two, proving principality. Thus no new root oracle is needed.

Pad the remaining prime axes by $n_i = p_i + 1$. Then $M/m' = \prod_i (1 + 1/p_i) = 1 + O(L_0^{-2})$. Since $\alpha' > L_0^{64}$, for sufficiently large N, $L = \lceil \log_2(2M) \rceil \leq L_0$ and $L = \Theta(L_0)$. The native radix hypotheses and $D' \geq P_1(R_0 + 1, L)$ hold together. Setup polynomial in R_0, D', L_0 is $N^{o(1)}$ and is charged once.

6 The coupled CRT index permutation

Write $m' = \prod_{i=1}^{t} p_i$, with least-significant-first digits

$$x = a_1 + p_1 a_2 + \cdots + (p_1 \cdots p_{t-1}) a_t.$$

Its physical lexicographic axis order is $[p_t] \cdots [p_1]$, already ordinary integer index order. In one lexicographic scan embed these records in the $[p_i + 1]$ grid, preserving their order, and zero-fill every address containing at least one dummy coordinate.

Lemma 6 (Parallel CRT tree)**.** *The permutation $a \mapsto (x \bmod p_i)_i$ and its inverse have cost $O(m' D' L_0^{\gamma} \operatorname{polylog} L_0)$.*

Proof. For a contiguous block $I = A \cup B$ with A lower significance, write $x_I = x_A + P_A x_B$, $P_A = \prod_{i \in A} p_i$. After converting both children to their local residues, the left residues are already correct for x_I. Each right target becomes

$$z_i \longmapsto P_A z_i + x_A \pmod{p_i}.$$

This is a permutation because P_A is a unit modulo p_i. The left residue address determines $x_A \in [0, P_A)$ by ordinary integer CRT arithmetic. Fix the target dummy, and use the identity when a relevant left control is dummy. This is an explicit bijection on every fiber and preserves every dummy-pattern class at completion.

At each level of a balanced tree take the union of all right axes as T, and all left/passive axes as C. The disjoint node merges are one controlled tensor from Lemma 4. There are $O(\log(t+1))$ levels. This schedule avoids two separate full-volume child executions per node. Reverse the stages and affine maps for the inverse. □

7 Characteristic-two local Fourier compilation

For an odd prime p, let $u \in S$ be a principal p-root. Every field component has exact order p, so $1 - u^j$ is a unit for $0 < j < p$. Set $H_j = \prod_{a=1}^{j} (1 - u^a)$ and $H_0 = 1$. The Newton matrix

$$N_{j,k} = \prod_{a=0}^{k-1} (u^j - u^a), \quad j \geq k,$$

is diagonal–lower-Toeplitz–diagonal, with $N_{j,k} = H_j H_{j-k}^{-1} (-1)^k u^{k(k-1)/2}$. The symmetric Newton factorization is $F_p(u) = N \operatorname{diag}(N_{kk})^{-1} N^T$. All pivots remain units over the finite etale algebra.

The explicit formal triadic circuit in Lemma 12, with one operand fixed, supplies a linear convolution DAG of size $O(b \log^8 b)$ and depth $O(\log^2 b)$ over every commutative $\mathbb{F}_2$-algebra. Bounded-degree copy trees are included. The local appendix also gives explicit constant/linear gate rules for this formal specialization, with no Boolean idempotence identity substituted in the coefficient ring.

The borrowed-coordinate mechanism of [6] restores arbitrary dirty workspace by replaying once with the true sources and once with the sources suppressed. Its algebra works in characteristic two. Inside a width-v Toeplitz recursion, choose chunk size $b = cv/\log^8 v$, with c a sufficiently small fixed constant. The two chunks and all gate coordinates fit within the same v positions. There are $O(\log^{16} v)$ source/target chunk pairs, each using $O(\log^2 v)$ replay rounds. A balanced recursion gives $O(\log^{19} p)$ rounds on exactly p coordinates, as proved in Lemma 13. Its source, target and borrowed ports are disjoint; output-copy wires

are counted inside the DAG. All borrowed values return before the next chunk pair. The finite small range is direct.

Work in $B = S \otimes_{\mathbb{F}_2} \mathbb{F}_4$ and take central $\eta^2 + \eta + 1 = 0$. Every coefficient $\mu \in B$ is the sum of two units: let e be one on the components where $\mu = 1$ and zero elsewhere, and put $d = 1+e(\eta+1)$. Then d and $\mu+d$ are units. All component degrees equal $r_B = \mathrm{lcm}(\mathrm{ord}_{\alpha'} 2, 2)$; thus $e = 1 + (\mu + 1)^{2^{r_B}-1}$ can be prepared by polynomially many small-algebra operations. For unit t,

$$U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = \mathrm{diag}(t, 1)HJ\,\mathrm{diag}(t^{-1}, 1), \qquad U(\mu) = U(d)U(\mu + d).$$

Here is one uniform slot word for the whole round. Pad each p axis by its one dummy and complete a partial shear matching by pairing its unused coordinates with coefficient $\mu = 0$. Order every used pair as (target,source), and send these pairs to canonical adjacent pairs by a local permutation. Set $u_0 = d$, $u_1 = \mu + d$ and $D(t) = \mathrm{diag}(t, 1)$. In rightmost-first matrix order,

$$U(\mu) = D(u_0)HJD(u_0^{-1}u_1)HJD(u_1^{-1}).$$

All diagonal entries are units. An unused pair has $u_0 = u_1 = 1$, so its completed word is $(HJ)^2 = I$; it need not be fixed at an intermediate letter. Tensoring uses exactly two full canonical H slots, two full J slots and three axis-factorized diagonal slots. The pair relabeling and its inverse each cost one generic tensor permutation from (1); charge the two J slots there as well. Thus a round has two native-H calls and at most four generic permutation calls. No separate selection of H versus I is needed on this local path. Each completed local axis map is $F_{p_i} \oplus [1]$ on its native-valid range $n_i = p_i + 1$. The tensor preserves each per-axis dummy-pattern subspace; it need not fix every full address containing a dummy, since other axes still transform. Consequently an initially zero-filled dummy complement stays zero at the completed tensor boundary. Keep the full padded grid through these operations, including every intermediate letter.

Lemma 7 (Independent runtime word arithmetic)**.** *There is a uniform fixed-tape algorithm for ordinary binary polynomial products at length v in $O(M_{\mathrm{old}}(v))$ time, with bound function*

$$M_{\mathrm{old}}(v) := v \log^2(2v) \log\log(4v).$$

It uses classical integer multiplication, independently of the proposed main construction and of the formal convolution circuit supplier.

Proof. For $v \geq 2$ set $k = \lceil \log_2(v + 1) \rceil$ and lift $\sum_{i<v} a_i X^i$ to $\sum_{i<v} a_i 2^{ki}$, with $a_i \in \{0, 1\}$. Every integer convolution coefficient is at most $v < 2^k$. Consequently the product's k-bit chunks are the exact integer coefficients, without carries between chunks. Their low bits give the ordinary $\mathbb{F}_2$ product. The encoding, zero padding, chunk extraction and optional bit-order reversals are sequential scans costing $O(vk)$. Counter resets are charged to their ensuing chunks. The classical integer algorithm of [4] takes $O(B \log(2B) \log\log(4B))$ steps on a fixed multitape machine for $B = vk$ bits, proving the bound. Marking visited work-tape intervals allows head returns and cleanup within the same order of time. Length one is direct. The bound function is stable under fixed-factor length changes. □

Tensor scalar coefficients are prepared by prefix expansion in the current native order, as detailed in the local appendix. The total number of products is below $2M$, not one product per axis at every record. A B symbol has $2D'$ bits. Each B multiplication or fixed-scalar application costs $O(M_{\mathrm{old}}(2D')) = O(M_{\mathrm{old}}(D')) = D' \,\mathrm{polylog}\, L_0$ bit steps using Lemma 7 and the explicit fixed-modulus reduction in the local appendix. In contrast, the routing

compiler's $\mathbb{F}_4$ scalars are fixed two-bit maps costing $O(D')$. These two costs must not be conflated. Project back to S only after the complete Fourier tensor has returned to that subspace. Each synchronized replay round has a fixed number of calls at the generic/native γ tariff and a fixed number of these scalar stages. There are $O(\log^{19} R) = \operatorname{polylog} L_0$ rounds, so the local Fourier tensor retains the same γ exponent.

8 Ordinary multiplication and the time ledger

Use Lemma 5 to construct $N = \Theta(n) > 2n - 2$ and its certified factor list, then enlarge the divisor as above. Initial input scans cost $O(n)$ and all host initialization is included in the $N^{o(1)}$ setup term below. The top Agarwal–Cooley isomorphism from [1, Lemma 2.2] is

$$X \longmapsto Y^c Z, \quad c = (m')^{-1} \bmod \alpha',$$

from $\mathbb{F}_2[X]/(X^N - 1)$ to $\mathbb{F}_2[Y, Z]/(Y^{\alpha'} - 1, Z^{m'} - 1)$. Its coprime factors α', m' both exceed two for sufficiently large n: α' tends to infinity and $\log \alpha' = o(\log N)$. It uses one classical rectangle transpose and cyclic row shifts in $O(N \log \alpha') = N \operatorname{polylog} L_0$ bit time, in either direction. Explicitly, an exponent $k = r + m'j$ goes to (Z-index, Y-index) $= (r, cr + j \bmod \alpha')$; this fixes the row-shift convention and its inverse. Since α' is odd, $\Phi_{\alpha'}$ is coprime to $\psi = (Y^{\alpha'} - 1)/\Phi_{\alpha'}$. Put $f = \Phi_{\alpha'}$ and $d = \deg \psi = \alpha' - D'$. If $d = 0$ the coefficient split is the identity. Otherwise use Lemma 14 on each α'-bit row for both projections, including the possibly much shorter modulus ψ. The latter's prepared reciprocal has length $\alpha' - d = D'$. Choose fixed Bezout polynomials $uf + v\psi = 1$, with $\deg u < d$ and $\deg v < D'$. The idempotents $e_f = v\psi$ and $e_\psi = uf$ have degree $< \alpha'$, and reconstruct

$$A = (A_f e_f + A_\psi e_\psi) \bmod (Y^{\alpha'} - 1).$$

Two padded length-α' word products and a linear cyclic fold suffice for this reconstruction. Projections and reconstruction thus cost $O(M_{\text{old}}(\alpha'))$ per row, and

$$O(m' M_{\text{old}}(\alpha')) = N \operatorname{polylog} L_0$$

overall. All reciprocals, Bezout coefficients and idempotents are public polynomial-α' setup, charged once. The two projected streams have total length $m'(D' + d) = N$ bits.

For the large component let ω be the principal m' root and

$$E_i = (m'/p_i)((m'/p_i)^{-1} \bmod p_i), \quad \omega_i = \omega^{E_i}.$$

The CRT idempotents give $\omega^{xy} = \prod_i \omega_i^{(x \bmod p_i)(y \bmod p_i)}$. Lemma 6 and the local Fourier tensor therefore compute the true ordinary cyclic DFT. Two forward transforms, point products and one inverse compute the product in $S[Z]/(Z^{m'} - 1)$. Retain the full native grid through these completed operations; its dummy complement remains zero at each such boundary and under the point product, so it can be cropped after inverse routing. The inverse uses inverse roots and reversed index routing; its odd normalization is one in characteristic two.

Here the large coefficient ring obtained by projection is $\mathbb{F}_2[Y]/\Phi_{\alpha'}(Y)$, identified with S by renaming Y to U. This identification is distinct from the earlier setup homomorphism $Y_{\text{old}} \mapsto U^J$ from the smaller ring $\mathbb{F}_2[Y_{\text{old}}]/\Phi_\alpha(Y_{\text{old}})$. That homomorphism constructs the root; it is not applied a second time to the new input words. In this section ω denotes the already constructed ω'. Indeed if $J > 1$, U^J has order α, rather than α', in every field component of S. Hence $\Phi_{\alpha'}(U^J)$ is a unit, so that repeated substitution would not even descend from the new quotient.

Frequency order needs its own verification. If local roots instead have exponents $b_i h_i$, with $b_i = m'/p_i$ and h_i a unit modulo p_i, an output digit s_i describes frequency residue $b_i h_i s_i$. The idempotent choice $h_i = b_i^{-1}$ makes that residue s_i. For a coordinate permutation Q, however, $A = QF$ still satisfies $A^{-1}((Aa) \odot (Ab)) = F^{-1}((Fa) \odot (Fb))$. Thus matching inverse and product checks alone do not certify a literal Fourier index convention.

The complement has $d = \deg \psi = \alpha' - D'$ and $v = dm' = N\theta = O(N\ell^2/L_0^3)$ by (2). If $d = 0$, omit this component. Pack $Y^j Z^i \mapsto T^{2di+j}$ in $\mathbb{F}_2[T]$. There is no overlap between coefficient-product slots because $j + j' < 2d$. Each input has degree at most $2v - d - 1$ and the product degree at most $4v - 2d - 2$. Use Lemma 7 at input length $2v$. Crop its known zero tail to $2d(2m' - 1)$ coefficient slots and unpack $2m' - 1$ rows of $2d$ bits, each with final bit zero. Reduce their degree-$\leq 2d - 2$ polynomials modulo the fixed monic ψ using Lemma 14, then fold $Z^{m'} = 1$. All packing, unpacking and folding scans cost $O(v)$; when $d = 1$ the row remainders are direct. The total cost is $O(M_{\mathrm{old}}(2v) + m' M_{\mathrm{old}}(d)) = O(M_{\mathrm{old}}(v) + m' M_{\mathrm{old}}(d))$. Since $d \leq v \leq N$, each logarithmic factor is bounded using L_0, so the total is

$$O(vL_0^2\ell) = O(N\ell^3/L_0) = o(N).$$

The scans and all reductions are included. The weaker estimate $v < N/L_0$ alone would not pay for this integer-packing fallback; the stronger density is essential here. No integer packing of the full ordinary input is used.

Lemma 8 (Typed top-level execution)**.** *The preceding component algorithms compose on a fixed number of streams, with ordinary coefficient input and output. Their complete calls use the stated tariffs; no coefficient projection or outer native-dummy crop occurs inside a Fourier word.*

Proof. Use the following order, with a, b denoting the two inputs.

1. Count and copy the coefficient lists, construct the host and all length-dependent tables, and zero-fill both lists to N coefficients. The live top-level input streams contain $2N$ bits. Their cost is $O(n) + N^{o(1)}$ before the linear zero-fill.

2. Apply the top isomorphism to each list, producing m' consecutive α'-bit coefficient rows. Project each row to the f and ψ components. The two large streams contain m' words of D' bits; the two small streams contain m' words of d bits. Park the small streams on a fixed separate pair of tapes. The maps and projections cost N polylog L_0. The large words are identified with S by the variable rename just specified.

3. Embed each large word in $B = S[\eta]/(\eta^2 + \eta + 1)$ as $a + 0\eta$, and zero-fill its native dummy grid. Keep these two M-word B arrays through both forward transforms, their point product and the inverse. They occupy $4MD' = O(N)$ bits at these top-level boundaries. Transform one while the other waits on its own tape; constant whole-array passing and parking scans cost $O(N)$. No callee traverses the parked array.

4. To specify every completed transform, let $E : B^{m'} \to B^M$ be the zero-fill embedding from integer mixed-radix order. Let Q be the completed extended CRT permutation, and put
$$\mathcal{T}_\omega = \bigotimes_i (F_{p_i}(\omega^{E_i}) \oplus [1]), \qquad \mathcal{D}_\omega = Q^{-1}\mathcal{T}_\omega Q.$$
The CRT exponent identity gives
$$\mathcal{D}_\omega E = E F_{m'}(\omega). \tag{3}$$

Thus the output is again in ordinary integer frequency order, in the same native grid. The whole operators Q and $\mathcal{T}_\omega$ preserve every dummy-pattern subspace. Their intermediate words may mix the auxiliary coefficient lanes or dummy values; only completed boundaries are used in this identity. Their valid block has entries in S, so a complete transform returns S-valued words when its input was S-valued. Keep the unused η lane allocated nonetheless.

5. Replace the first completed transform array by the coordinatewise B product of the two arrays, scanning them together, and erase the second array. On valid coordinates this is exactly the S product; on every dummy coordinate it is zero. Performing all M products, without an exceptional-address lookup, costs $O(MM_{\mathrm{old}}(D')) = N \operatorname{polylog} L_0$. Apply $\mathcal{D}_{\omega^{-1}}$ to the first array. The principal-root character sums imply $F_{m'}(\omega^{-1})F_{m'}(\omega) = m'I = I$. Equation (3) and the convolution identity therefore give the cyclic product in $S[Z]/(Z^{m'} - 1)$. Explicitly, for the projected coefficient vectors a_f, b_f the jth inverse coefficient is

$$\sum_{r,s=0}^{m'-1} (a_f)_r (b_f)_s \sum_{k=0}^{m'-1} \omega^{k(r+s-j)} = \sum_{r+s\equiv j \pmod{m'}} (a_f)_r (b_f)_s.$$

The inner character sum is zero off the stated congruence and is $m' = 1$ on it. Thus the complete retained-grid computation is $\mathcal{D}_{\omega^{-1}}((\mathcal{D}_\omega E a_f) \odot (\mathcal{D}_\omega E b_f)) = E(a_f *_{\mathrm{cyc}} b_f)$. Only now remove the zero η lane and crop the zero dummy complement, using monotone native-address scans costing $O(N)$. The result is m' consecutive D'-bit words in ordinary Z order. Generate the dummy test once per record using the fixed degree-twelve address procedure; its cost is paid by $D' \geq P_1$.

6. Compute the parked small-component product by the independent packing and fixed-monic reductions above, at cost $o(N)$, obtaining m' consecutive d-bit words, omitting this branch if $d = 0$. Reconstruct the α'-bit rows from the two components and apply the inverse top isomorphism at cost $N \operatorname{polylog} L_0$. Extract the first $2n - 1$ coefficients in a final $O(N)$ scan. All other top-level streams are erased and their heads returned by scans charged to their lengths.

There are three complete transforms, each with a constant number of CRT maps and the stated synchronized local rounds. Their total is $O(NL_0^\gamma \operatorname{polylog} L_0)$, since their B payload width is $2D'$ and hence satisfies the same metadata lower bound. The number of top-level streams is fixed; setup archives and recursive scratch use the separately charged tape families, rather than one fresh tape per axis, table or recursion level. □

The full-volume routing and Fourier arrays have $V = \Theta(N)$ and address width $L = \Theta(L_0)$. The smaller complement arrays have the separate budget above. The stage counts are fixed powers of $\log L_0$. The complete bound furnished by the construction is

$$O(NL_0^\gamma (\log L_0)^C) + N \operatorname{polylog} L_0 + N^{o(1)}.$$

With $\epsilon = 2^{-36}$, the reconstructed interchange and phase certificates give $\tau = 1 - \epsilon$ and $\sigma < \tau$. Hence

$$\gamma = \frac{1}{1+\epsilon}, \qquad 1 - \gamma = \frac{\epsilon}{1+\epsilon} > 2^{-37}.$$

For fixed C, all secondary logarithms are therefore absorbed in $O(n(\log n)^{1-2^{-38}})$. The same exponent is retained through the controlled compiler and CRT tree; their extra work is linear metadata and fixed powers of $\log L_0$, not a larger power of L_0. The effective finite initial

range uses a fixed direct branch. Reconstruction and inverse top routing return the ordinary cyclic N coefficients; degree $2n-2 < N$ means the first $2n-1$ coefficients are exactly the ordinary dense polynomial product.

Order of choices and one cutoff. The finite motifs, avoiding prime, generator degrees and exponents are fixed first. Host construction then fixes N, L_0, ℓ, R_0 before enlargement. The threshold for D' uses L_0, and the resulting native width satisfies $L \le L_0$; it is not fed back into the host. Only $O(\ell)$ extra factors are consumed from $\Omega(L_0/\ell^2)$ available factors. All setup degrees and secondary logarithmic powers are fixed constants of these algorithms. The imported host constants are effective [1, Section 1]; the other constants come from the printed finite generators and constructive bounds. One maximum of their finitely many eventual thresholds therefore supplies the direct-branch cutoff. For explicit final logarithm absorption, write $\delta = 2^{-38}$ and bound the main term by $AN L_0^{\gamma}(\ln L_0)^C$ with fixed $A \ge 1$. For

$$t = \ln L_0 \ge \max\{1, (2C/\delta)^2, 2\ln(A)/\delta\},$$

$\ln t \le \sqrt{t}$ gives $A(\ln L_0)^C \le L_0^{\delta}$. Since $1-\gamma > 2\delta$, this fits the stated $1-\delta$ exponent. No input-dependent exponent or length advice is selected.

Theorem 9 (Main theorem)**.** *There is one uniform deterministic machine with a fixed finite alphabet and a fixed finite number of one-dimensional tapes which multiplies two ordinary dense degree-$<$ n polynomials over $\mathbb{F}_2$, for every $n \ge 1$, and returns their $2n-1$ ordinary coefficients in time*

$$O\Big(n(\max\{1, \lceil \log_2 n \rceil\})^{1-2^{-38}}\Big).$$

Its construction uses no unproved prime-distribution hypothesis.

Proof. Lemma 5 supplies the actual admissible length, factor list, divisor and root from n in charged $N^{o(1)}$ setup, with $N = \Theta(n) > 2n-2$. Word enlargement keeps $m'D' = \Theta(N)$ and makes all controlled-routing coefficients computable within their word cost. The reconstructed interchange, finite phase and native endpoint appendices prove Proposition 1; the ragged and controlled compilers therefore price the actual index permutations. Lemma 6 supplies the ordinary-to-prime-residue index map. The local appendix gives the exact prime Fourier words over the specified finite etale algebra, with separately charged classical scalar maps. The preceding component calculation computes the large and complementary coefficient-ring products, reconstructs the cyclic N-product, and extracts the ordinary product. The displayed time ledger absorbs every fixed secondary logarithm in the stated exponent. Lemma 8 specifies the actual stream composition, root/coefficient types and the single final auxiliary-lane/dummy crop. The finite tape and pad composition below applies to these actual subroutine calls. A direct finite branch handles the bounded initial range, with one enlarged uniform constant. □

Corollary 10 (Each fixed finite extension of $\mathbb{F}_2$)**.** *For every fixed integer $s \ge 1$, multiplication of ordinary dense degree-$<$ n polynomials over $\mathbb{F}_{2^s}$ can be performed by one deterministic machine with a fixed finite alphabet and fixed finite number of tapes, in running time*

$$O_s\Big(n(\max\{1, \lceil \log_2 n \rceil\})^{1-2^{-38}}\Big).$$

This is a derived corollary of Theorem 9; its constants and machine may depend on s.

Proof. Fix a basis of $\mathbb{F}_{2^s}$ over $\mathbb{F}_2$ and write each input coefficient in that basis. By bilinearity, each output coordinate of the product is a fixed $\mathbb{F}_2$-linear combination of the s^2 ordinary binary coefficient products of the input coordinate polynomials. Run Theorem 9 on those s^2

pairs in sequence, reusing the same finite tape family, and combine the resulting coefficient lists using the fixed multiplication table of the chosen basis. The s^2 calls cost $O_s(n(\log(n+2))^{1-2^{-38}})$; the coefficientwise recombination costs $O_s(n)$. For each fixed s this gives the stated bound. No uniformity as s grows with n is asserted. □

Tape and padding composition. Host I/O, classical integer SS, ring arithmetic, metadata generation, generic routing, H2 roles, weighted tiles and SWAP have disjoint fixed tape families. Each recursive family reuses its tapes with current-parent stack parking. No child traverses parked ancestors. The caller waits while a subroutine operates on its actual stream; every passing/return scan is included in that subroutine's cost. The native dummy grid is kept through complete routing/Fourier operations. Private rectangle, tile, role and suffix pads have the precise lifetimes in the machine ledger; none is repeatedly multiplied into recursive child mass. The complete family call graph, head-return rules and crop table are specified in the included machine-composition record `supporting/STRICT_BINARY_MACHINE_COMPOSITION_ROOT_20261030.txt`, Sections III–V.

Verification evidence and remaining review. A previously recorded ragged-network witness checked 42,090 permutations at even sizes at most 12. A previously recorded guarded C++ witness checked 768 cases on native axes $(4, 6, 8)$, including all 105 valid CRT addresses for primes $(3, 5, 7)$, inverse maps and exact dummy patterns. Omitting the control gauge fails. Both install 64 MiB/10 CPU-second limits before algebra. They validate finite identities and conventions only. The bundled machine-composition derivation and proof map document the integration dependencies; internal audits do not replace external mathematical review or establish novelty. These existing witness reports were not rerun for this release copy. A previously recorded guarded fixed-array C++ witness checked all 65 536 coefficients of the small product algebra $(\mathbb{F}_2[Y]/\Phi_{15}) \otimes \mathbb{F}_4$ for the selector/unit split and the local two-H word, with independent bilinear multiplication checks. Previously recorded mixed-radix tensor fixtures checked scalar-prefix expansion, arbitrary incoming values and dummy restoration. Wrong middle factors and stale scalar alignment are rejected. This remains bounded algebraic evidence, not a test of the asymptotic tape bound. A previously recorded bounded whole-composition witness used $N = 255 = 17 \cdot 15$, both coefficient CRT components, the affine residue-index tree, compiled 3- and 5-point Newton words and ordinary reconstruction. It checks the literal forward/inverse DFT on all 15 basis vectors and 96 ordinary binary products through degree bound $n = 128$. Omitting row shifts, the complement or the required extension lifetime is detected. Consistently changing primitive roots preserves its roundtrips/products while failing the literal Fourier comparison. The witness uses direct finite primitive branches and does not test the growing-size running-time theorem. A previously recorded guarded C++ witness checked the elementary triadic product over $\mathbb{F}_2[\epsilon]/(\epsilon^2)$: 130 ordinary products through length 512, a detected wrong-overlap control and a nilpotent-law check. It uses direct finite DFTs and base cutoff 32 to exercise the quotient-product recursion. The proof's radix-three depth and analytic cutoff 6^6 are separate; this witness establishes neither the asymptotic circuit bounds nor the main theorem. An expanded source-hashed finite check also verified the 776-letter E word, the explicit (A.17) conjugation and monomial scalar ordering. Ordinary-squaring and stale-scalar negative controls are detected. Its fixed four-by-four field and 16-bit binary maps do not test the growing-size native routing or time bound.

A Finite ternary phase router and a positive binary H_2 word

The scalar bank-exchange and frame-label architecture follows the finite-network method of [5]; its characteristic-two and ternary-phase adaptation is made explicit below. These finite-network antecedents are credited independently of the present tape-time claim.

This appendix records the finite algebraic and rank-saving part of the binary H_2 supplier. It distinguishes three coefficient/address domains: the address-frame geometry is over $\mathbb{F}_3$, the phase kernel and dirty-selector payload arithmetic are over $\mathbb{F}_4$, and the fixed positive word is over its codec field $\mathbb{F}_{16}$. The tape costs for the equal-chunk interchange and weighted native mask are separate machine suppliers; no field element in these algebraic arguments is treated as a unit-cost operation on a growing coefficient word.

A.1 The ternary phase kernel

Let $\eta \in \mathbb{F}_4$ satisfy $\eta^2 + \eta + 1 = 0$. On $\mathbb{F}_4^3$ put

$$C = \eta I_3 + \eta^2 J_3 = \begin{pmatrix} 1 & \eta^2 & \eta^2 \\ \eta^2 & 1 & \eta^2 \\ \eta^2 & \eta^2 & 1 \end{pmatrix}.$$

Then $C^3 = I$. Let $\chi(a) = \eta^a$ for $a \in \mathbb{F}_3$, and let

$$(\mathcal{F}_3)_{x,z} = \chi(xz), \qquad x, z \in \mathbb{F}_3.$$

Since $1 + \eta^a + \eta^{2a} = 0$ for nonzero $a \in \mathbb{F}_3$, $\mathcal{F}_3^2$ is the coordinate-negation permutation. Thus $\mathcal{F}_3^4 = I$ and $\mathcal{F}_3^{-1} = \mathcal{F}_3^3$ (equivalently, $(\mathcal{F}_3^{-1})_{x,z} = \eta^{-xz}$). Direct multiplication gives

$$C = \mathcal{F}_3 \operatorname{diag}(1, \eta, \eta)\, \mathcal{F}_3^{-1}.$$

Thus the eigenphase of C at frequency z is η^{z^2}. The inverse is C^2 and has eigenphase η^{-z^2}.

For an address space with a nondegenerate symmetric form over $\mathbb{F}_3$, write $T_v f(x) = f(x + v)$. If v has norm $\nu = \langle v, v \rangle \in \{1, 2\}$, define

$$K_v = I + \eta^{-\nu}(T_v + T_{-v}). \tag{A.1}$$

On a Fourier character of frequency z, the two translations have eigenvalues $\eta^{\langle v,z\rangle}$ and $\eta^{-\langle v,z\rangle}$. For $a = \langle v, z\rangle \in \mathbb{F}_3$,

$$1 + \eta^{-\nu}(\eta^a + \eta^{-a}) = \eta^{a^2/\nu}. \tag{A.2}$$

Indeed both sides are 1 for $a = 0$; for $a = \pm 1$, checking $(\nu, a^2) = (1, 1)$ and $(2, 1)$ gives η and η^2 respectively. Consequently K_v is precisely the one-direction quadratic phase for the line $\langle v\rangle$. Orthogonal directions commute, and an orthogonal direct sum of directions has the product of their phases. Replacing v by $-v$ does not change the line; norm 2 replaces C by C^{-1}.

For a nondegenerate subspace U with an orthogonal basis $v_1, \ldots, v_r$, define $C_U = \prod_j K_{v_j}$. In Fourier coordinates this is the multiplier

$$(\mathcal{F} C_U \mathcal{F}^{-1})_{z,z} = \eta^{\sum_j \langle v_j, z\rangle^2 / \langle v_j, v_j\rangle}. \tag{A.3a}$$

Thus orthogonal direct sums multiply their C_U operators. If $U \subseteq V$ and $V = U \perp W$, then $C_V = C_U C_W$; a rank-r edge between such frames needs exactly r one-direction children. The phase frame on a role is a proof invariant: the same $\mathbb{F}_4$-linear phase operator is applied to aligned wire arrays, so it commutes with pointwise XOR/addition. It is a linear operator

on payload arrays, not an address relabeling; this is distinct from the address permutations used in the SWAP supplier. At completion the role permutation pairs each input frame with its specified output frame. Choosing the X-source inverse line frame makes the endpoint product the full phase, while its first edge has difference $P_U - (-P_U) = 2P_U$ and therefore one rank-one child. This is the source correction counted below.

A.2 The 48-triple frame over $\mathbb{F}_3$

Fix $H = 48$, let $\mathcal{T} = \binom{[H]}{3}$, and join distinct S, T when $|S \cap T| = 1$. Put

$$v = \binom{48}{3} = 17296, \quad z = 3\binom{45}{2} = 2970, \quad N = v^3, \quad I = 3v^2.$$

The factor space is $F = \mathbb{F}_3^{49}$ with form

$$B = \operatorname{diag}(-I_{48}, 1).$$

For $S \in \mathcal{T}$ use $t_S = (1_S, 1)$. Then

$$\langle t_S, t_S \rangle = -3 + 1 = -2 = 1 \pmod 3, \qquad \langle t_S, t_T \rangle = 1 - |S \cap T|.$$

Thus the line $L_S = \langle t_S \rangle$ is unit norm and adjacent triples give orthogonal lines. The form is nondegenerate. To identify it with standard dot-product coordinates, split the first 48 coordinates into 24 disjoint pairs, and on each pair use

$$(a, b) \longmapsto (2a + 2b,\ 2a + b).$$

The two output vectors for the basis inputs are $(2, 2)$ and $(2, 1)$; their norms are 2 and 2, and their dot product is $2 \cdot 2 + 2 \cdot 1 = 0$ in $\mathbb{F}_3$. The map is invertible (determinant 1 in $\mathbb{F}_3$), so this is an isometry from $\operatorname{diag}(-1, -1)$ to the standard dot product. The last coordinate already has norm one.

For a stage prefix tensor space $A = F^{\otimes(j-1)}$, let P be its prefix line, $B_0 = P^{\perp}$, and let L_t denote the current triple line. Since P and L_t are nondegenerate,

$$A = B_0 \perp P, \qquad F = L_t \perp L_t^{\perp},$$

and

$$(P \otimes L_t)^{\perp} = (B_0 \otimes F) \perp (P \otimes L_t^{\perp}). \tag{A.3}$$

The neighbor condition supplies $L_s \perp L_t$ whenever a side role contains the ordered neighbor (s, t). These decompositions prove the following eight gate labels are nested/comparable exactly where their wire edges require:

$$\begin{array}{l|l} t_0 & B_0 \otimes L_t \\ t_1 & B_0 \otimes F \\ t_2 & (B_0 \otimes F) \perp (P \otimes L_t) \\ t_3 & A \otimes F \\ t_4 & B_0 \otimes F \\ t_5 & (B_0 \otimes F) \perp (P \otimes L_t^{\perp}) \\ t_6 & A \otimes F \\ t_7 & A \otimes F. \end{array} \tag{A.4}$$

In stage two the physical wire order is the inverse of the forward row schedule with logical source and target exchanged; this keeps the same physical X/Y label convention in (A.4). In

particular, its incoming physical X label is $(A \otimes L_t) \otimes Q$ and its outgoing label $(A \otimes F) \otimes Q$; its incoming physical Y label is $(B_0 \otimes L_t) \otimes Q$ and outgoing label $(P \otimes L_t)^{\perp} \otimes Q$. The latter is (A.3) for the next stage, with the suffix tensor line Q restored. This checks the interstage edges as well as the within-invocation inclusions.

Here is the wire-level interpretation of the table. A stage fixes the other two triple indices of a data role in $\mathcal{T}^3$; all displayed labels are tensored with its fixed suffix line Q. Name an ordered neighbor pair by its physical Y and physical X triple indices in that stage, renaming the ordered pairs in the inverse second stage. The side wire $a_{(S,T)}$ meets, in order, the Y-side gate for S, the X-side gate for T, the Y-side gate for S, and the X-side gate for T. Their labels are rows t_0, t_2, t_5, t_7. Its middle inclusion is precisely $L_T \subset L_S^{\perp}$, supplied by the neighbor condition. Each center wire meets t_1, t_3, t_4, t_6, whose labels alternate $B_0 \otimes F$, $A \otimes F$, $B_0 \otimes F$, $A \otimes F$. Data X wires meet t_2, t_3, t_6, t_7 and data Y wires meet t_0, t_1, t_4, t_5. These descriptions check every gate incidence; multi-input sums can be decomposed into the same common-label XOR gates without adding frame edges of nonzero rank. Side and center scratch wires have identity input phase and full-space output phase, connecting their first/last displayed labels to zero/full labels. Initial Y data have identity phase. Final X data have full-space phase and final Y data have the complement of their triple tensor line. The initial X inverse line is the sole exceptional source frame. The physical scratch rows also require the full phase: they were arbitrary rows of the original array. It is their logical scalar scratch values that the routing motif restores before the frame is interpreted at the output.

For a fixed eligible mask sector, let P be this scalar role permutation acting on aligned arrays, and let $D_{\text{in}}, D_{\text{out}}$ be the block-diagonal input/output phase frames. A frame change on an edge from F to G applies GF^{-1}. Common-label XOR gates commute with their shared phase, so the entire framed program telescopes to $D_{\text{out}} P D_{\text{in}}^{-1}$. If input role r is sent to $P(r)$, the endpoint operator there is $C_{\text{out},P(r)} C_{\text{in},r}^{-1} = C_{\text{full}}$ for data *and* scratch roles. Consequently

$$D_{\text{out}} P D_{\text{in}}^{-1} = P(I_W \otimes C_{\text{full}}). \tag{A.4a}$$

The inverse role merge removes P, leaving the full phase on every arbitrary input role array. Eligibility is the same mask-only predicate on every row role; in an ineligible sector every frame is identity, so the scalar P is likewise canceled by that merge. No scratch payload is initialized to zero in this identity.

A.3 Arbitrary-scratch scalar routing and the finite rank sum

For one invocation over $\mathbb{F}_2$, let x be its data source, y its target, a its side scratch vector and c its center scratch vector. Let V, J, G, R be the incidence maps in the companion scalar-motif certificate. The chronological eight rows are

$$\begin{aligned} y &\leftarrow y - Ja, & y &\leftarrow y - Rc, \\ a &\leftarrow a + Vx, & c &\leftarrow c + Gx, \\ y &\leftarrow y + Rc, & y &\leftarrow y + Ja, \\ c &\leftarrow c - Gx, & a &\leftarrow a - Vx. \end{aligned}$$

where in the first two rows a, c are their entry values. In characteristic two subtraction equals addition, but the signs make the scratch cancellation visible. The final two rows restore the center and side scratch. Equivalently, for arbitrary initial scratch (a_0, c_0) the total target increment is

$$-Ja_0 - Rc_0 + R(c_0 + Gx) + J(a_0 + Vx) = (RG + JV)x. \tag{A.5}$$

For triples, $G_{i,T} = 1_{i \in T}$, $R_{S,i} = 1_{i \in S}$, $V_{(S,T),T'} = 1_{T=T'} 1_{|S \cap T|=1}$ and $J_{S,(S,T)} = 1_{|S \cap T|=1}$. If $S = T$, $RG = 3 = 1$ in $\mathbb{F}_2$ and $JV = 0$. If $S \neq T$ with intersection one, both terms are one

and cancel; for intersections zero or two both vanish. Hence $RG + JV = I$. Formula (A.5) proves exact addition and restoration for every scratch assignment.

Run the forward schedule X-to-Y, its inverse schedule Y-to-X, then the forward schedule again. On arbitrary banks

$$(x, y) \mapsto (x, y + x) \mapsto (-y, y + x) \mapsto (-y, x), \tag{A.6}$$

so in characteristic two this exchanges the banks. Every invocation has disjoint scratch roles; all are fixed by the completed role permutation. There are v roles per bank, vz ordered side roles and H center roles per invocation, hence

$$W = 2N + I(vz + 48). \tag{A.7}$$

For a label of dimension r over $\mathbb{F}_3$, a wire operation on that label uses its quadratic phase and a source/target frame change of rank equal to the relevant dimension difference. The only downward label transitions in (A.4) are the $t_3 \to t_4$ center transitions. Each loses $F = 49$ dimensions and there are 48 center coordinates in each of I invocations. Their total downward loss is

$$L = I \cdot 48 \cdot 49. \tag{A.8}$$

The signed dimension telescope over all routed wire endpoints is $WM - 2N$, where $M = 49^3 = 117649$: the two banks contribute the two N endpoint deficits. Turning signed variation into absolute variation adds twice every downward loss, so before the source correction the sum of edge ranks is $WM - 2N + 2L$.

This count can also be read from the complete physical wire paths. Put $d = 49$ and $a = d^{j-1}$ at stage j. With the suffix line suppressed, the side and center dimensions are respectively

$$0,\ a-1,\ d(a-1)+1,\ da-1,\ da,\ M, \qquad 0,\ d(a-1),\ da,\ d(a-1),\ da,\ M.$$

The side inclusions follow from $L_T \subset L_S^{\perp}$, with nondegenerate rank-$(d-2)$ residual. The center loses exactly d dimensions at $t_3 \to t_4$ and otherwise increases. Physical X enters the next stage as $A \otimes L_T \subset t_2(T)$; physical Y enters as $B_0 \otimes L_S = t_0(S)$. Their complete ordinary paths have no decreases and have total rank $M - 1$ each, using a formal positive X input line for this count. The inverse middle stage retains this physical order: its inverse logical rows $7, \dots, 0$, after bank exchange, have the same physical incidences as chronological rows $0, \dots, 7$. The label table itself is not reversed. Consequently the total ordinary rank is

$$2N(M-1) + (W - 2N)M + 2I \cdot 48d = WM - 2N + 2L.$$

Choose the X-source phase frame to be the inverse line phase on U_a; all other source frames are identity and gate/sink frames are the positive phases of (A.4). The endpoint identity on each X-to-Y route is $C_{U_a^{\perp}} C_{U_a} = C_{F^{\otimes 3}}$. On each of the N first X edges, the source label is $-P_{U_a}$ while the first gate label is $+P_{U_a}$, so their difference is $2P_{U_a}$. This has rank one and its phase is $C_{U_a}^{-1}$, since scaling the quadratic form by $2 = -1$ in $\mathbb{F}_3$ inverts its phase. These are N genuine extra children, not a rank-zero equal-dimension edge. Therefore

$$s = WM - 2N + 2L + N = WM - N + 2L. \tag{A.9}$$

At $H = 48$ direct cancellation gives

$$\frac{W}{N} = \frac{9633881}{1081}, \qquad \frac{L}{N} = \frac{441}{1081}, \qquad 1 - \frac{s}{WM} = \frac{199}{9633881 \cdot 117649}. \tag{A.10}$$

The integer comparison

$$199\,2^{36} > 12 \cdot 9633881 \cdot 117649$$

and $\log_2 M < 17$, $\ln 2 < 7/10$ imply $\ln M < 12$ and

$$\frac{199}{9633881 \cdot 117649} > 12\,2^{-36} > 1 - M^{-2^{-36}}.$$

Thus $s/W < M^{1-2^{-36}}$. This is the explicit strict role saving used below; no numerical matrix-rank test is involved.

A.4 From the role word to all active sectors

An A-digit has states $0, 1, 2$ active and state 3 idle. Represent $d_i = 2X_i + Y_i$. For any active mask S, $A^{\otimes h}$ restricts to $C^{\otimes|S|}$ on the active coordinates and identity on idle coordinates. All 4^h addresses remain in the stream; no sector is gathered out.

The mask-routing identities use dirty address fields. Let T_Z apply one fixed transposition of active symbols where dirty selector bit $Z_i = 1$. It preserves the active mask S, but arbitrary dirty Z can make it change paired sources as well as targets. Source-value predicates are recomputed only after the completed $T_{J\wedge Q}$ word below, which changes matched targets only. For a mask-only selector $J(S)$ the chronological word

$$Z \oplus\!= J(S), \quad T_Z, \quad Z \oplus\!= J(S), \quad T_Z \tag{A.11}$$

restores arbitrary initial Z and equals T_J. With a second dirty field Q, use $J(S) \wedge Q$ in (A.11). If $F_a(d)_i$ says the paired source of target i currently has trit a, then the completed $T_{J\wedge Q}$ changes targets only and preserves all paired sources. Consequently

$$Q \oplus\!= F_a(d), \quad T_{J\wedge Q}, \quad Q \oplus\!= F_a(d), \quad T_{J\wedge Q} \tag{A.12}$$

restores arbitrary Q, Z and implements the target transposition conditioned on source value a. The source/target matching must be disjoint; this is what makes the second F_a equal to the first at the completed inner-operation boundary. A third dirty field is reserved for the weighted tile auxiliary. The supplied weighted-mask lemma implements each frozen XOR in (A.11)–(A.12) with its separate auxiliary restored; its exact sparse boundary repair and interchange cost are not reproved here.

The two active transpositions have explicit bit circuits

$$(0\ 1):\ Y \oplus\!= Z \wedge \neg X, \qquad (1\ 2):\ X \oplus\!= Z \wedge Y; \quad Y \oplus\!= Z \wedge X; \quad X \oplus\!= Z \wedge Y. \tag{A.13}$$

They are constant sequences of frozen-control XOR shears. Applying the two transpositions in chronological order (1 2) then (0 1) adds one to an active trit; reverse order subtracts one. Conditioning on source trit and composing these shears realizes a matching SUM

$$\text{target}_i \leftarrow \text{target}_i + \text{source}_i \pmod 3. \tag{A.14}$$

Three SUMs plus a negation implement a coordinate swap. All dirty selectors and the tile auxiliary return to their arbitrary incoming values after each completed operation.

For a node acting on the first e active ranks, restrict controls and endpoints to the first

$$u = \min\{h, \lceil C_0(e + \log(L+2))\rceil\}$$

physical positions, where $L \geq h$ is the ambient address-width bound in the native implementation. If fewer than e active positions occur there, define the operation to be identity;

otherwise all required ranks are present. Restrict every raw T_Z transposition and both selector-bank toggles to this prefix, not just the completed selector J. Insert inherited eligibility into $J(S)$; it is not a frozen control on an inner bit shear whose target can change S temporarily. Individual inner shears still read only the other bit plane and the selector bank. Their completed transposition preserves S. The comparison with the full intended route is made only after a *complete logical* mask-selected transposition, source-conditioned operation (A.12), SUM or recursive basis placement on the first e active ranks, with all its dirty selectors restored. Write this prefix implementation as P_u and its intended permutation as T. On a good eligible prefix all intended endpoints are inside it, so $P_u = T$. On a bad or ineligible prefix the selector J is zero: the two raw transpositions in (A.11) cancel, each inner operation is identity, and the two source-predicate toggles in (A.12) cancel too. Thus the completed P_u is identity there. These statements do not require raw dirty T_Z to preserve paired sources.

One must not repair an internal truncated T_Z to its full-h version: with $h = 2, u = e = 1$, $d = (0, 0)$ and $Z = (0, 1)$, the prefix is good but full T_Z for (0 1) changes the second digit and truncated T_Z does not. This toy example embeds for any actual $u < h$ by putting the dirty selector immediately after the prefix and using active digits throughout the prefix. That off-prefix action is intended to cancel in the outer word. Nor is the prefix bad set invariant after an inner bit shear: the first shear of (1 2) sends idle $(X, Y) = (1, 1)$ to active $(0, 1)$. Its tile repair uses a different invariant, as specified in the native appendix. The separate top-level digit-group placement can reach ranks beyond e; it retains its full-h root-level charge and is not justified by this width-e prefix argument. In the uniform $[4]^h$ address space the prefix active count is binomial with mean $3u/4$. For $u \geq 2e$ and $u < h$, a Chernoff bound gives

$$\Pr\{B_u < e\} \leq e^{-u/24}. \tag{A.15}$$

For $u = h$, eligibility $|S| \geq e$ leaves no exceptional eligible mask. Both T and the completed logical P_u preserve S, so the exceptional set $E_{\text{inherited}}(S) \wedge [|S| \geq e] \wedge [B_u < e]$ is invariant. Its whole-universe fraction is bounded by the unconditional binomial estimate; no conditional distribution given eligibility is assumed. At that completed boundary, sorting the exceptional records by $T(P_u^{-1}(y))$ and refilling their slots repairs it exactly; the maps agree everywhere else. No sector is dropped. The full role recursion always stores the complete $[4]^h$ address universe, and its row split depends only on the independent role field, never on S. Thus every child retains the same mask distribution and off-eligibility maps are identity. Consequently, on every sector S, the role program applies $C^{\otimes |S|}$ to its active digits and identity to its idle digits; the direct sum over all masks is exactly $A^{\otimes h}$.

The prefix correction above belongs to the completed logical routing permutation, including T and T^{-1} below, after all its source conditioning and before a linear combination forms a phase. An internal $T_{J \wedge Q}$ is not separately corrected when its surrounding source predicate was prefix-truncated too. It never sorts the output of the non-permutation matrix C.

At selected width one, let T cyclically permute that eligible active trit and be identity on ineligible masks. The mask-only transpositions above implement T and T^{-1}. Three temporary array copies and a pointwise linear combination compute $I + \eta^2(T + T^{-1})$, which is C on the selected trit. Off eligibility the two identical terms cancel, leaving the identity. The copies, returns and cleanup add $O(V)$ at this base call. Norm two uses its inverse via the fixed coefficient wrapper described below.

The $M = 49^3$-slot basis changes in the finite frame program decompose into a fixed number of row-addition matching batches on the first e selected ranks. Nondegenerate residual spaces over $\mathbb{F}_3$ have orthogonal bases: repeatedly split a nonisotropic line; its orthogonal complement remains nondegenerate. The sign of the directed frame difference must also be retained.

Lemma 11 (Signed direction and one-child implementation)**.** *Let v be one such direction in standard-dot coordinates, with $\nu = \langle v, v\rangle \in \{1, 2\}$, and let $\varepsilon \in \{1, 2\}$ specify the positive or negative directed phase. The effective child coefficient is*

$$\lambda = \varepsilon/\nu \quad \text{in } \mathbb{F}_3.$$

For the block direction $\langle v\rangle \otimes \mathbb{F}_3^f$, use one ordinary width-f child if $\lambda = 1$, and one conjugated inverse child if $\lambda = 2$. Both cases retain exactly the same child volume.

Proof. Choose $A \in \mathrm{SL}_M(\mathbb{F}_3)$ with $Av = e_1$. Extend v to a basis and adjust the sign of another column to obtain determinant one; $M \geq 2$. A fixed elementary row-addition word realizes A. Define the address pushforward by $(Q_A f)(y) = f(A^{-1}y)$. Its translation conjugacy is $Q_A^{-1} T_w Q_A = T_{A^{-1}w}$. Chronologically place the addresses by $A \otimes I_f$, apply the first-block child $(C^{\otimes f})^\lambda$, and undo the placement. The resulting operator is

$$Q_{A\otimes I_f}^{-1} (C^{\otimes f})^\lambda Q_{A\otimes I_f}.$$

Its Fourier multiplier is

$$\eta^{\lambda \sum_{\ell=1}^f \langle v, z^{(\ell)}\rangle^2}.$$

The desired directed line phase has exponent

$$\varepsilon \sum_\ell \langle v, z^{(\ell)}\rangle^2/\nu,$$

so the stated λ is exact. The placement need not preserve the quadratic form: ν is computed before placement, in the original form. Replacing v by $-v$ leaves these squares unchanged.

For $\lambda = 2$, wrap the *same forward child* with the coefficientwise $\mathbb{F}_4$ Frobenius on both existing codec planes, as in (A.17). This gives $(C^{\otimes f})^{-1}$ on arbitrary codec values. The wrapper is $\mathbb{F}_2$-linear; its completed conjugate is linear over the codec field. It adds two linear child-stream scans, and no second recursive call or new coordinate. If eligibility is false, the child is identity and the two involutions cancel there. □

Explicitly, the forward/inverse cases are

ε	ν	λ	child
1	1	1	forward
1	2	2	inverse
2	1	2	inverse
2	2	1	forward

An upward nested-frame edge has $\varepsilon = 1$; a downward edge, including $t_3 \to t_4$, has $\varepsilon = 2$ on its residual subspace. The exceptional source difference splits as a positive residual projection plus $2P_{U_a}$, so its extra unit-line child in (A.9) also has $\lambda = 2$. This flag selects an operator, not an extra child; the rank sum s is unchanged.

At selected width $e = Mf$, each rank-a edge uses exactly a width-f children, one for each orthogonal basis direction. With s the total child count (A.9), and W roles, normalized time obeys

$$F(e; h) \leq (s/W) F(e/M; h) + O(\mathrm{polylog}(L + 2)(e^\tau + \mathrm{polylog}(L + 2))). \tag{A.16}$$

The additive term includes dirty-selector shears, prefix exception repairs, basis placements and current-parent tape parking. Put $\sigma = \log_M(s/W)$. Exact unrolling gives

$$O(e^{\max\{\sigma,\tau\}} \mathrm{polylog}(L + 2)).$$

Equality $\sigma = \tau$ adds only $\log e = O(\log L)$. Here the reconstructed interchange permits $\tau = 1 - 2^{-36}$ and $\sigma < \tau$, so the ratio $(s/W)/M^{\tau}$ is strictly below one and the bound is

$$O(e^{\tau} \operatorname{polylog}(L+2)).$$

No additional exponent slack is required at this step. Arbitrary active counts are partitioned into base-M power-width groups; digit-conditioned groups partition each mask's active ranks, and each top-level placement is performed once per group, not again in every child. Hence the complete $A^{\otimes h}$ tensor has a strict exponent in h, relative to the stated interchange and weighted-mask suppliers.

A.5 Codec field and explicit positive word for $H_2 \otimes H_2$

The address field $\mathbb{F}_3$ above is not the phase field. For the constant word use

$$E = \mathbb{F}_2[t]/(t^4 + t + 1), \qquad \eta = t^5 = t^2 + t.$$

The quartic has no root in $\mathbb{F}_2$ and remainder 1 modulo the only irreducible quadratic t^2+t+1, so it is irreducible. Thus $E = \mathbb{F}_{16}$, it contains $\mathbb{F}_4 = \mathbb{F}_2(\eta)$, and $\eta^2 + \eta + 1 = 0$. The four-state phase letter is

$$A = (\eta I_3 + \eta^2 J_3) \oplus [1] = C \oplus [1].$$

It acts separately on the two $\mathbb{F}_4$ coefficient planes in $E = \mathbb{F}_4 \oplus t\mathbb{F}_4$. Its inverse is A^2. The inverse-phase wrapper on these planes is

$$\widehat{\sigma}(a + tb) = a^2 + tb^2 \quad (a, b \in \mathbb{F}_4), \tag{A.17}$$

which applies the $\mathbb{F}_4$ Frobenius to each plane and fixes the codec basis element t. It is $\mathbb{F}_2$-linear and is not ordinary $\mathbb{F}_{16}$ squaring. In the polynomial basis $(1, t, t^2, t^3)$ it is the explicit map

$$(c_0, c_1, c_2, c_3) \longmapsto (c_0 + c_2 + c_3,\ c_1 + c_3,\ c_2,\ c_3).$$

Indeed $a_0 = c_0$, $a_1 = c_2 + c_3$, $b_0 = c_1 + c_2 + c_3$ and $b_1 = c_3$ in $a = a_0 + a_1\eta$, $b = b_0 + b_1\eta$. Three in-place XORs implement it, and repeating them gives the identity. For $\lambda \in \mathbb{F}_4$, $\widehat{\sigma}(\lambda z) = \lambda^2 \widehat{\sigma}(z)$. Consequently coefficientwise conjugation gives

$$\widehat{\sigma} C \widehat{\sigma} = \eta^2 I_3 + \eta J_3 = C^2 = C^{-1}, \qquad \widehat{\sigma}(C^{\otimes f})\widehat{\sigma} = (C^{-1})^{\otimes f}.$$

This holds on arbitrary E-valued inputs. The two passes cancel on idle or ineligible coordinates; either pass alone can change their coefficient values. Both passes act within the existing four-bit lanes, in $O(V)$ time around one child. Ordinary E squaring used on both sides would fail even on an idle value t, sending it to $t^4 = t+1$ instead of t. The semilinearity assertion is only for $\mathbb{F}_4$ scalars, not arbitrary E scalars.

Here is the explicit word construction. Put $a = \eta^2$ and $b = \eta$; use column vectors and rightmost-first composition. Then

$$A^3 = I, \quad A^{-1} = A^2, \quad u = (1, a, a, 0)^T, \quad v = (1, b, b, 0)^T, \quad P = AE_{11}A^{-1} = uv^T,$$

with $v^T u = 1$, hence $P^2 = P$. Let

$$D = \operatorname{diag}(t, 1, t+1, 1), \quad Q = DPD^{-1} = x_0 y_0^T,$$

where $x_0 = (t, a, a(t+1), 0)^T$ and $y_0 = (t^{-1}, b, b/(t+1), 0)^T$. Direct products give

$$v^T x_0 = 0, \qquad y_0^T u = \kappa = \frac{t^2 + t + 1}{t(t+1)} \neq 0, \qquad PQ = 0, \quad QP = \kappa x_0 v^T. \tag{A.18}$$

For idempotents P, Q with $PQ = 0$,

$$[I + cP, I + dQ] = I + \frac{cd}{1+c} QP. \tag{A.19}$$

This follows by using $(I + cP)^{-1} = I + c(1 + c)^{-1}P$ and $QPQ = 0$. With $c = d = \eta^2$, $cd/(1 + c) = 1$. Thus

$$\begin{aligned} G &= A \operatorname{diag}(\eta, 1, 1, 1) A^{-1} = I + \eta^2 P, \\ H &= DGD^{-1} = I + \eta^2 Q, \\ T_0 &= GHG^{-1}H^{-1} = I + \kappa x_0 v^T. \end{aligned} \tag{A.20}$$

Each of G, H, G^{-1}, H^{-1} uses three forward A letters after replacing inverse A by A^2, so T_0 uses 12.

The remaining conjugations isolate an elementary matrix. Set

$$B = \operatorname{diag}(1, \eta, \eta, 1), \quad x = (t, 1, t + 1, 0)^T, \quad w = (1, 1, 1, 0)^T.$$

Then $T_1 = BT_0B^{-1} = I + \kappa x w^T$ (12 A letters). Let P_{12} swap coordinates 1 and 2. Since w is fixed and $w^T P_{12} x = 0$,

$$T_2 = T_1 P_{12} T_1 P_{12}^{-1} = I + \kappa(t + 1)(e_1 + e_2)w^T \tag{A.21}$$

uses 24 A letters. With $D_3 = \operatorname{diag}(1, 1, \eta, 1)$,

$$T_3 = T_2 D_3 T_2 D_3^{-1} = I + \kappa(t + 1)\eta(e_1 + e_2)e_3^T \tag{A.22}$$

uses 48; the two cross pairings vanish and the row sum is $(1 + \eta^{-1})e_3^T = \eta e_3^T$. With $D_1 = \operatorname{diag}(\eta, 1, 1, 1)$,

$$T_4 = T_3 D_1 T_3 D_1^{-1} = I + \kappa(t + 1)E_{13} \tag{A.23}$$

uses 96; the cross pairings vanish and the column factor is $1 + \eta = \eta^2$. Put $\mu = \kappa(t + 1) \neq 0$; then

$$T_5 = \operatorname{diag}(\mu^{-1}, 1, 1, 1) T_4 \operatorname{diag}(\mu, 1, 1, 1) = I + E_{13}. \tag{A.24}$$

All conjugating factors are invertible monomial matrices. Coordinate permutations produce every $U_{ij} = I + E_{ij}$ with the same 96 forward A letters. This is an explicit formula, not an uncomputed word search.

Finally write $H_2 = US$, with $U = I + E_{12}$ and S the two-coordinate swap. In ordered coordinates $(00, 01, 10, 11)$, put

$$P_{\text{first}} = (1\ 3)(2\ 4), \qquad P_{\text{second}} = (1\ 2)(3\ 4).$$

Then, using the four explicit elementary-matrix words,

$$H_2 \otimes H_2 = U_{13} U_{24} P_{\text{first}} U_{12} U_{34} P_{\text{second}}. \tag{A.25}$$

There are four U_{ij} factors, hence $4 \cdot 96 = 384$ forward A letters; the remaining factors expand to 392 invertible monomial letters, for 776 total. The completed matrix has entries in $\mathbb{F}_2$. As an $\mathbb{F}_2$-linear map on the four binary coordinates of an E-symbol, it applies $H_2 \otimes H_2$ independently to each coordinate plane. Thus a binary 0/1 embedding returns to 0/1, and any zero lane used for padding returns to zero at the completed word. An arbitrary E-valued input need not return to the scalar subfield. Intermediate E lanes must not be cropped. When four arbitrary payload bits are packed into one E symbol, all four coefficient planes remain data; only extra zero bits from the final partial group are removed. No projection $E \to \mathbb{F}_2$ is applied to such a packed symbol.

Pairing $2h$ identical native endpoint axes and tensoring this fixed word reduces their algebraic action to $A^{\otimes h}$ letters and fixed monomial letters. The permutation part of each monomial is in $\mathrm{AGL}(2,\mathbb{F}_2) = S_4$: its NOT/CNOT address shears and the product of its h diagonal E scalars are charged by the separately supplied native mask/routing and record-scan interfaces. The active-sector role construction reduces the $A^{\otimes h}$ letters. The native large-even-radix endpoint appendix supplies their tape price. No binary 2^k-axis H2 bound is asserted here; that requires a distinct binary mask-routing argument and is not needed for the manuscript's native endpoint interface. For precision, write a monomial as $M_{\pi(d),d} = m_d$, with $m_d \in E^*$. Apply the coefficient $\prod_i m_{d_i}$ at its current *source* address, then route by $\pi^{\otimes h}$. Equivalently an output-address scan would use $\prod_i m_{\pi^{-1}(e_i)}$. Using the source-indexed list unchanged after routing is generally incorrect. Tensoring the fixed word is letterwise: $(L_1 \cdots L_{776})^{\otimes h} = L_1^{\otimes h} \cdots L_{776}^{\otimes h}$. Thus there are 384 tensor phase calls, not 384 separate calls per axis.

Scope. The mask shears used here are constructed independently in Section B.2; interchange is proved in Appendix D. The mask lemma never calls the later native H2 tensor theorem. Conversely the rational interchange proof uses only the scalar incidence motif and finite subspace inclusions, not this phase recursion. These cross-references follow an acyclic dependency order, despite the order in which appendices are printed.

Equations (A.5)–(A.10) give the finite scratch-safe routing and strict rank saving; (A.11)–(A.16) give the active/idle and dirty-selector interface with inherited mask eligibility; (A.17)–(A.25) give the explicit positive word. Their tape implementation is relative to the separately proved weighted-native-mask and equal-chunk interchange suppliers. The appendix does not reprove those tape suppliers, the ordinary polynomial multiplication composition, or external acceptance of the claimed asymptotic result.

B Canonical endpoint kernels on mixed-radix tape addresses

This appendix supplies a strict-saving implementation for the canonical adjacent-pair kernel on a product of even native radices. It does not implement arbitrary pair matchings, a cyclic Fourier index permutation, or ordinary multiplication by itself. The finite phase-word supplier and the equal-chunk interchange contract are stated explicitly below; their component proofs are cited separately.

B.1 Model, assumptions, and supplied finite phase word

Let the address set be

$$\Omega = \prod_{i=1}^{t} [n_i], \qquad n_i \text{ even}, \qquad M = \prod_i n_i, \qquad L = \lceil \log_2(2M) \rceil,$$

with

$$L^3 \le n_i \le R, \qquad \log R = O((\log L)^c)$$

for a fixed constant c. Each record has $D \ge 1$ arbitrary payload bits. The machine has a fixed finite alphabet, a fixed finite number of one-dimensional tapes, and ordinary binary shape descriptors. All address work, parking, padding, and record motion are charged. Write V for the complete current stored bit volume, including padded address fields and opaque record lanes; on the original unpadded stream $V = MD$.

We use the interchange construction in Appendix D, based on the finite-network method of Section 4 of the September 2026 source *Integer multiplication below* $n \log n$: two u-bit

address fields in

$$[P] \times [2^u] \times [G] \times [2^u] \times [B]$$

can be interchanged in $O(Vu^\tau)$ bit steps with $\tau = 1 - 2^{-36} < 1$, where V is the complete stored bit volume. This choice follows from our exact 48-triple reconstruction, not from attributing that numerical exponent to the source preprint. The prefix, gap, and suffix are arbitrary, including opaque payload. This appendix uses only this interchange procedure, not the paper's global integer multiplication theorem.[1]

The second finite supplier is the exact word

$$A = (C \oplus [1]), \qquad C = \eta I_3 + \eta^2 J_3, \qquad \eta^2 + \eta + 1 = 0 \text{ in } \mathbb{F}_4.$$

The certificate proves that a fixed word in forward A and invertible monomial 4-by-4 matrices over $\mathbb{F}_{16}$ equals $H_2 \otimes H_2$, where

$$H_2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_2.$$

It has 384 forward A letters and 392 monomial letters. This is the finite phase-H supplier used here; no generic or unproved H_2 tensor theorem is assumed. Appendix A supplies the finite word and role proof; the earlier detailed word certificate is in `supporting/ntt_base_five/H2_FINAL_FINITE_WORD_CERTIFICATE.md`. The priced recursive tensor supplier for $A = C \oplus [1]$ is the finite ternary role/frame construction of `supporting/H2_FINAL_TERNARY_ROLE_CERTIFICATE.md`, with its dirty-router interface detailed in `supporting/H2_TENSOR_CLOSED_PROOF.md`. The role construction is used only through the finite constants, frame identities, and recurrence restated below. The mixed-radix endpoint masks in this appendix replace its binary mask router.

B.2 Weighted two-bank tile mask

For an integer pair (x, w), define the four-update word $P(a, b, s)$ by the chronological operations

$$x \mathrel{+}= [w \equiv b \pmod 2], \quad w \mathrel{+}= s[x \equiv a \pmod 2],$$
$$x \mathrel{-}= [w \equiv b \pmod 2], \quad w \mathrel{-}= s[x \equiv a \pmod 2].$$

The parity tests are evaluated at that step. Set $p = P(1, 1, +1)$ and $q = P(1, 0, -1)$. A direct check on the four parity tiles gives $p = (B\ C\ D)$ and $q = (A\ B\ D)$, with $A = (0, 0), B = (1, 0), C = (0, 1), D = (1, 1)$. Thus the chronological word p^{-1} followed by q is

$$(x, w) \longmapsto (2\lfloor x/2 \rfloor + (1 - (x \bmod 2)), w).$$

For a frozen control bit g, mask every one of these eight elementary updates by g. The resulting word is identity for $g = 0$ and the displayed bit flip for $g = 1$. In particular it restores the whole integer w, not merely its parity. This is the binary tile identity in `supporting/ntt_base_five/binary_packed_affine_monomial_routing.md`.

Now take u digit pairs with even radices n_i, m_i, all at least L^3, and ordinary numeric orders

$$X = \sum_{i=0}^{u-1} x_i A_i, \quad A_i = \prod_{j<i} n_j; \qquad W = \sum_{i=0}^{u-1} w_i B_i, \quad B_i = \prod_{j<i} m_j.$$

Each g_i is computed solely from frozen address controls disjoint from X, W and the dock below; it cannot depend on a bank while that bank is being updated. In one X-update of

[1] Primary statement: Section 4, equal-chunk interchange. The finite role and tape reconstruction dependencies are listed at the end of this appendix.

the eight-step word, let $\delta_i \in \{-1, 0, 1\}$ be its local signed increment, determined only by $w_i \bmod 2$, g_i, and the public word position. Execute the bank update by the single weighted integer increment

$$\Delta_X = \sum_i A_i \delta_i.$$

For a W-update, define ϵ_i from $x_i \bmod 2$ and set

$$\Delta_W = \sum_i B_i \epsilon_i.$$

The first offset is independent of X on each fixed (W, controls) fiber; the second is independent of W on each fixed (X, controls) fiber. Hence each update is a cyclic line rotation of its target numeric bank, not a gather of its component digits.

To implement it in the fixed-tape model, reserve a separate binary dock Z of width $c_0 = \max(a, b)$, where $a = \lceil \log_2 \prod_i n_i \rceil$ and $b = \lceil \log_2 \prod_i m_i \rceil$. Pad each target range once to 2^a or 2^b (factor below two), retaining its true validity predicate. For an X-update, interchange its a-bit field with an a-bit slice of Z using the supplied equal-chunk theorem. On the valid prefix $0 \le X < \prod n_i$, rotate modulo the true range by Δ_X; copy the invalid tail pointwise. If W is invalid, use offset zero on the valid X prefix. Interchange back. The W-update is symmetric, forcing zero offset when X is invalid. Every exchange restores arbitrary dirty dock contents. Consequently every privately padded state is fixed pointwise by each elementary update and may be cropped only after the completed word. No unequal-side transpose across an arbitrary gap is presumed. Validity follows the *logical target*: after docking X, test its value in the dock, not the old physical X position, which now contains arbitrary dock data. The other bank remains in place. The internal numerical pads of each SWAP call have already been cropped when that call returns, before this valid-prefix rotation.

Put $N_X = \prod n_i$, $N_W = \prod m_i$. Suppressing Cartesian spectators, the original native domain is $\Omega_0 = [N_X] \times [N_W]$ and the privately padded domain is $\widetilde{\Omega} = [2^a] \times [2^b]$. Every complete docked update fixes $\mathcal{I} = \widetilde{\Omega} \setminus \Omega_0$ pointwise. For the concrete schedule below keep these private pads through the repair, then crop them after the exact mask is complete.

On Ω_0, define the good set G_0 by quotient margins

$$10 \le \lfloor x_i/2 \rfloor \le n_i/2 - 11, \qquad 10 \le \lfloor w_i/2 \rfloor \le m_i/2 - 11 \quad \text{for all } i.$$

Each local integer moves by at most four during the eight updates, so on this set no digit overflows and the weighted updates equal the independent tile words. The ideal completed map toggles only x_i's endpoint bit; it preserves every quotient label and fixes W. It therefore preserves G_0 and $\mathcal{B} = \Omega_0 \setminus G_0$ separately. Extend the ideal map as identity on $\mathcal{I}$. The packed word P agrees with that extension on $G_0 \cup \mathcal{I}$, and both are permutations of $\widetilde{\Omega}$. Hence $P(\mathcal{B}) = T(\mathcal{B}) = \mathcal{B}$. The union bound gives density on the *original native domain*

$$\delta = \frac{|\mathcal{B}|}{|\Omega_0|} \le 40 \sum_i (n_i^{-1} + m_i^{-1}) = O(u/L^3).$$

The repair marker is explicitly

$$[X < N_X] \wedge [W < N_W] \wedge [\text{a quotient margin fails}].$$

It is false on $\mathcal{I}$, even if private padding occupies a constant fraction of $\widetilde{\Omega}$. Do not decode an invalid numeric bank as a wrapped native digit tuple. Extract only marked records at their current padded output addresses y. Recover their sources $x = P^{-1}(y)$ by reversing the eight packed updates, then compute the ideal endpoint destination $T(x)$. These are public address calculations with frozen controls; no old key is assumed to accompany the

payload. Keys are destination ordinals in the same current padded layout. Stably sort these destination keys and refill the same bad slots. This corrects every exceptional address and discards no payload. The repair costs $O(\delta V L + M_{\mathrm{rec}} L^C)$ for a fixed C, where M_{rec} is the native record count before these private bank pads. The padded record count and bit volume are each below four times their native values, so full marking/refill scans cost $O(V)$. For clarity, with $K \le \delta M_{\mathrm{rec}}$ bad records, use stable binary radix sorting on the $b = O(L)$ key bits. In each pass copy one fixed-width key/payload item to a staging tape, remember the selected bit, rewind, and copy to the appropriate zero/one bucket. Concatenate the buckets and rewind for the next pass. A fixed number of tapes and $O(K(R_{\mathrm{op}} + b))$ steps suffice per pass, including all item and file returns. Counter updates are amortized over the scanned length, and resetting a counter costs at most the following item or block length. The bound is therefore $O(KL(R_{\mathrm{op}} + L))$, plus explicitly charged polynomial address work and full-array marking/refill scans. At refill, consume the next sorted item exactly at a marked bad slot; $P(\mathcal{B}) = T(\mathcal{B}) = \mathcal{B}$ proves the order without a full-key comparison at every good record. The validity-and-margin predicate above marks $\mathcal{B}$ directly, so no extra mask lane is added to the retained field codec. The exponential opaque width below absorbs the key-only and metadata terms. The tile word uses a constant number of dock interchanges and rotations, hence costs $O(V(u \log R)^\tau + V)$. In the application below, the bundled opaque record width is exponential in L, so it absorbs $M_{\mathrm{rec}} L^C$; also $\delta L = O(u/L^2) = O(1)$. The resulting mask and its inverse cost

$$O(V(u \log R)^\tau + V).$$

After refill the padded map is exactly T on Ω_0 and identity on $\mathcal{I}$, with the dock restored. Crop only the private X, W numeric tails now, returning the original native ranges before any enclosing phase child. A caller's Fourier dummy p_i in a native range $n_i = p_i + 1$ is *inside* that range and is not such a tail; it remains part of the native kernel. Role-row, suffix and coefficient bit pads also retain their separate outer crop boundaries below.

B.3 Native pair endpoints and disjoint banks

For a native digit $x_i \in [n_i]$, write $x_i = 2a_i + b_i$ with $b_i \in \{0, 1\}$. The desired local matrix applies H_2 to $\{2a_i, 2a_i + 1\}$ for each fixed a_i. Equivalently, for every fixed tuple $(a_1, \ldots, a_t)$ it applies $H_2^{\otimes t}$ to the endpoint tuple $(b_1, \ldots, b_t)$, while the physical interleaved mixed-radix record order remains unchanged. This is a statement about fibers of the native address set; it does not perform or price a tensor regrouping or an endpoint gather. The labels a_i are spectators. Because every n_i is even, endpoint bits are independent uniform bits under uniform native addresses, and pairs of endpoint bits from disjoint axes are uniform four-state digits. No arbitrary matching is hidden in this factorization.

For a large block of $2h$ endpoint axes, use the already contiguous product rank of native intervals; no per-record mixed-radix conversion is performed. One aggregate rectangle exchange moves the chosen interval to the front. For adjacent ranges $[A][B]$, pad once to $A' < 2A, B' < 2B$, perform the binary rectangle transpose, and crop the completed rectangle before returning. Its volume is below four times the original and its charged cost is $O(VL^\gamma)$, where

$$\gamma = \frac{1}{2 - \tau} < 1.$$

This follows from Euclidean decomposition into equal-chunk interchanges; partial internal exchanges need not preserve the valid rectangle. Reverse the complete transpose after the block. This is a top-level layout cost, not a cost paid in each recursive role child.

Choose

$$\ell_R = \lceil \log_2 R \rceil, \qquad h = \min(\lfloor t/16 \rfloor, \lfloor L/(40\ell_R) \rfloor).$$

For sufficiently large L, $h \geq 1$ and $5h < t/2$. The first two disjoint h-axis banks are X, Y. From the remaining consecutive native interval take three more disjoint h-axis banks Q, Z_0, W; Z_0 here is an endpoint bit bank, distinct from the private binary dock Z. The other native axes form a spectator suffix. The F16 word works on four-state digits $d_j = 2b_j + b_{h+j}$, pairing the jth axis of X with the jth axis of Y; Q, Z_0, W are dirty endpoint-selector banks. This is logical bank indexing only: the bits remain in their two contiguous native banks, and each shear is executed by the mask oracle without physically interleaving X and Y. All controls for one weighted mask are chosen from the other disjoint banks and role bits, never its target, tile auxiliary, or dock.

More explicitly, the dirty-router shears for $(0\ 1) : Y \oplus= Z_0 \wedge \neg X$ use X, Z_0 as frozen controls. The three shears for $(1\ 2) : X \oplus= Z_0 \wedge Y$, $Y \oplus= Z_0 \wedge X$, $X \oplus= Z_0 \wedge Y$ each use the other plane and Z_0. The outer source-value selector F_a for Q is computed from X, Y; the inner selector for Z_0 is $J(S)$ or $J(S) \wedge Q$, computed from X, Y, Q. Each is reversed after use at its specified completed boundary. The tile auxiliary for every call is a slice of the separate native W bank, disjoint from the current target and every control. Role bits may be added as frozen controls. These are the control dependencies of the dirty-selector construction in `supporting/H2_TENSOR_CLOSED_PROOF.md`, Lemma 2; no selector is recomputed from a bank while that bank is the weighted tile auxiliary.

Pad the numeric range of the whole native spectator suffix once to its next power of two, a factor below two. Split this already contiguous binary index into high role bits, then dock bits, then low bits. Write r for the number of role bits, s for the padded suffix width, and

$$c_0 = \max_{G \in \{X,Y,Q,Z_0,W\}} \lceil \log_2 |G| \rceil \leq h \log R + 1$$

for the reserved dock width. The high row field is borrowed from these existing suffix index bits, not added as another Cartesian factor. Reserve there at least $W_*^{\lceil \log_m h \rceil}$ rows, padded once by a factor below two as in the finite role supplier; each recursive level divides this field by W_*. Follow it by the c_0 dock bits. The remaining low suffix index is bundled with the external payload. Since $5h \log R \leq L/8$, the suffix has width at least $7L/8 - O(1)$; $r = O(\log h)$ and $c_0 \leq L/40 + 1$. Thus for all sufficiently large L the opaque record width obeys

$$R_{\mathrm{op}} = 2^{s-r-c_0} D \geq 2^{4L/5} D.$$

Move only the r high role bits to the front by individual binary digit moves, costing $O(Vr) = O(V \log h)$, and undo these moves once the complete block returns. The dock is now the rightmost active address field, immediately before the low opaque lanes. For a maximum role depth k, choose r with $2^r \geq W_*^k$, and pad this one independent row range to $W_*^k \lceil 2^r / W_*^k \rceil < 2^{r+1}$. The row pad stays outside the role recursion. In the program used here it is retained through the entire fixed H block, including all phase and monomial letters, and is cropped after that completed word restores row labels. Monomial letters act only on X, Y and codec payload, so they preserve this independent numeric row coordinate. Here r counts borrowed binary suffix bits *before* this private padding. After padding, the row range is a separate numeric factor R', which need not fit in r bits. Its counter can use $r+1$ bits; this is metadata, not another bit borrowed from the low opaque index. The array volume grows by $R'/2^r < 2$, already included in the current V. Role splitting uses numeric rows and SWAP permits an arbitrary numeric prefix, so neither requires an r-bit encoding of R'. Crop the row range back to its original 2^r values before undoing the r binary digit moves. Thus no genuine suffix states are dropped, and the exponent $s - r - c_0$ for the original low opaque lanes is unchanged. All polynomial-in-L selector generation and repair metadata is paid by this width, even when $D = 1$. No control reads the low suffix index or the original payload.

B.4 A fixed metadata program and sufficient suffix threshold

Here is a concrete bound for the instruction family actually used, rather than an allowance for every possible polynomial-time function. Decode the five bank indices by at most $5L$ schoolbook divisions on L-bit integers, at cost $O(L^3)$. Store the active-position list, pair matching and any group-placement permutation in at most L records of $O(L)$ bits each. Charge $O(L^2)$ for a full-scan lookup; there is no unit-cost indexed access. A fixed-slot basis instruction joins ranks $(i-1)f+a$ and $(j-1)f+a$, for $a=1,\dots,f$, with distinct fixed slots i,j. A group placement is an interval rotation of the active list; its cycle decomposition uses at most L seeds, L steps per seed and full-scan lookup/marking. The two cycle reflections then give its disjoint swap lists.

The recursive descriptor retains a list of $O(\log h)$ mask-only eligibility conditions, not an expanded recursive Boolean expression. They are conjunctions of active-count, prefix and base-m digit tests; all completed coordinate-value maps preserve the physical active mask. Every actual Boolean-vector instruction is one of $F_a(X,Y)$, $J(S)$, $J(S)\wedge Q$, $Z_0\wedge Y$, $Z_0\wedge X$, $Z_0\wedge\neg X$, or a fixed NOT/CNOT. A constant number of the same routines computes each weighted offset and each inverse packed update used for $T(P^{-1}(y))$. Field scalars are products of at most h entries from fixed $\mathbb{F}_{16}$ tables. These are fixed finite compositions of loops with bounds at most L or L^2. Allowing redundant list copies and descriptor scans gives the deliberately loose bound $O(L^{16})$ time and $O(L^4)$ metadata storage per current address. Copying the opaque word and moving sort payloads remain separate, explicitly linear-volume charges.

Fix $P_{\mathrm{nat}}(L)=L^{64}$. It pays this actual generator whenever $R_{\mathrm{op}}\ge L^{64}$. For an explicit suffix choice let $k=\lceil\log_m h\rceil$ and $r=56k$, where $m=117649>2^{16}$. The fixed phase role count below obeys $W_*<18000^3\cdot 9000<2^{56}$, so $2^r\ge W_*^k$ and $r\le 56+(7/2)\log_2 L$. Since $s\ge 7L/8-2$ and $c_0\le L/40+1$,

$$s-r-c_0\ge (17/20)L-(7/2)\log_2 L-59.$$

For $L\ge 2^{16}$ this is at least $(4/5)L$, and $2^{(4/5)L}\ge L^{64}$. Thus the original suffix supplies the fixed metadata allowance for every external $D\ge 1$. The privately padded numeric role range still adds less than two volume, without consuming another original suffix bit. This certificate concerns nonempty admissible five-bank blocks; the direct finite branch also covers smaller L and parameters for which $h=0$.

One may fix the prefix constant to 4096:

$$u=\min\{h,\lceil 4096(e+\lceil\log_2(L+2)\rceil)\rceil\}.$$

If $u<h$, the prefix exception fraction is at most $\exp(-u/24)\le (L+2)^{-128}$; if $u=h$, eligible masks have no prefix exception. More precisely the repaired mask is $E_{\mathrm{inherited}}(S)\wedge[|S|\ge e]\wedge[B_u<e]$; off-eligibility sectors already have the correct identity action and are not sent through a full-payload correction sort. Its exact permutation repair has $O(V)$ payload cost. The separate tile-boundary fraction is at most $80u/L^3$ and also gives $O(V)$ repair. These statements count addresses in the retained universe and leave no probability of an incorrect output. The two repairs have different boundaries. Tile repair follows one completed eight-update endpoint XOR and uses invariant quotient labels. Prefix repair follows a completed logical active-value route, with its source conditioning and dirty selectors restored; only that whole route agrees with its intended full-rank permutation on good prefixes. No prefix repair is applied to an internal raw T_Z or bit CNOT. The prefix implementation is identity on the exceptional masks, so its repair key there is simply the ideal logical destination. Both repairs precede the linear combination defining C and leave the retained codec and opaque width unchanged.

B.5 Finite phase recursion with inherited masks

Group corresponding endpoint bits into h four-state digits. State 3 is idle, and the active mask is $S = \{j : d_j \neq 3\}$. The finite ternary role supplier implements $A^{\otimes h}$, $A = C \oplus [1]$, by active-rank frames, dirty-selector commutators, and a fixed finite list of matching SUM batches. The finite frame counts are restated here to specify the exact recurrence. At ground-set parameter $g = 48$, put

$$\begin{aligned} v &= \binom{48}{3}, \quad z = 3\binom{45}{2}, \quad N = v^3, \quad I = 3v^2, \\ d &= 49, \quad m = d^3 = 117649, \\ W_* &= 2N + I(vz + 48), \quad L_* = I \cdot 48 \cdot 49, \\ s_* &= W_* m - N + 2L_*. \end{aligned}$$

There are W_* row-role streams and s_* recursive children. Every child has volume V/W_*. The source rank-one endpoint correction is included in s_*. The exact integer certificate gives

$$\begin{aligned} \frac{W_*}{N} &= \frac{9633881}{1081}, \qquad \frac{L_*}{N} = \frac{441}{1081}, \\ 1 - \frac{s_*}{W_* m} &= \frac{199}{9633881 \cdot 117649}, \\ \sigma &= \log_m(s_*/W_*) < 1 - 2^{-36} = \tau. \end{aligned}$$

Indeed $1 - s_*/(W_* m) = (N - 2L_*)/(W_* m)$. The finite certificate uses $m < 2^{17}$, hence $\ln m < 17 \ln 2 < 12$, and $199 \cdot 2^{36} > 12 \cdot 9633881 \cdot 117649$ to obtain the displayed strict exponent without a floating-point estimate. The H=48 phase, frame, endpoint, and source-count derivation is in `supporting/ntt_base_five/h2_from_chunk_swap_integrated_audit.md` (especially its Sections 3–4) and `supporting/ntt_base_five/finite_motif_rank_closed_form_audit.md`.

Prepared recursive entry. The recursive routine is PhasePower$(e, h, R_{\mathrm{row}}, B_E)$, with the bank/dock/codec descriptors and inherited mask eligibility also passed. The maximum native radix remains R; R_{row} denotes a distinct numeric row range. The entry requires $e = m^k$, with R_{row} divisible by W_*^k, and the already prepared five full h-digit native banks. Here $B_E = 4\lceil R_{\mathrm{op}}/4 \rceil$ is the existing opaque codec word, containing arbitrary E values. Its actual input volume is

$$V = R_{\mathrm{row}} B_5 2^{c_0} B_E,$$

where B_5 is the product of the five native bank ranges. An edge direction invokes

$$\mathrm{PhasePower}(e/m, h, R_{\mathrm{row}}/W_*, B_E)$$

directly after its basis placement. The child keeps the same physical h, bank ranges, dock, opaque width and codec; its row range is divisible by W_*^{k-1} and its volume is exactly V/W_*. This entry performs no bank layout, suffix padding, row padding, or new field embedding. Its return has the same prepared shape and can be nonbinary. For a directed residual phase with sign ε and direction norm ν, record $\lambda = \varepsilon/\nu \in \{1, 2\}$ as in Lemma 11. Use the ordinary child for $\lambda = 1$; $\lambda = 2$ wraps this one call on both existing $\mathbb{F}_4$ planes, with no additional child or payload-plane split. Thus downward unit-norm directions and the source correction use the inverse, while downward norm-two directions use the forward child. The explicit four-bit map after (A.17) is applied pointwise to the current child input and output, in two $O(V/W_*)$ scans. It changes no address, row, bank, dock, codec width, or parked parent

data. Conjugacy uses the completed child's $\mathbb{F}_4$ phase coefficients; the wrapper is not an automorphism for arbitrary E multiplication.

Only a root preparation routine constructs the independent row and codec. The suffix, row and conservative codec factors are respectively below two, below two, and at most four, so its prepared volume is at most $16MD$ after aggregate rectangle pads have been cropped. The routine for a full phase letter, PhaseAll, calls the exact power routine on its base-m groups. A power child does not re-enter PhaseAll or root preparation. Explicitly write the current active count as $a = \sum_j a_j m^j$. For each fixed j and $i = 0, \dots, m-2$, use eligibility $a_j > i$, group width $e = m^j$, and group start $\sum_{\ell<j} a_\ell m^\ell + i m^j$. Move that consecutive active group to the first e active ranks, call the prepared power entry, and undo its placement. For each mask these groups concatenate, without overlap, to its a active ranks. All passes keep the same numeric row and codec.

Each completed logical active-value route preserves S and restores its dirty auxiliaries. Inner bit shears need not preserve S; their separate tile correction uses quotient labels instead. A fixed monomial can change S, but it is itself an exact address permutation; recompute the active mask for the next role pass. The row-role split is independent of S. Hence every child retains the complete physical mask distribution and the same role multiplicity. Eligibility is a mask-only predicate with the active-count test at entry; off-eligible frames act as identity. Prefix concentration is applied to the current mask: for

$$u = \min\{h, \lceil C_0(e + \log(L+2))\rceil\},$$

the first u four-state digits contain at least e active ranks except with probability at most $\exp(-u/24)$, for a sufficiently large fixed C_0. The bad-prefix sector is extracted only at a completed logical route boundary, after all nested source conditioning. At that boundary the implemented route equals the ideal on good prefixes and is identity on the exceptional masks; its dirty selectors have returned. The ideal router preserves the whole active mask S, so this exceptional sector is invariant and sorting/refill by the ideal destination is exact. The weighted tile guard exception from the preceding subsection is repaired separately; neither correction crops or drops a record.

Induction on the full universe. For each physical mask S, the width-e contract is $C^{\otimes e}$ on its first e active ranks if the inherited predicate and $|S| \geq e$ hold, and identity otherwise. The child receives this complete eligibility test, including the parent's threshold; checking only $|S| \geq e/m$ would be insufficient. Fixed basis placements act on these active values and preserve S even when the selected ranks are not physically contiguous. By induction each edge direction has its specified signed phase on arbitrary E-valued payloads. For $\lambda = 2$ the two-plane Frobenius conjugation completes to the required E-linear inverse phase. The finite common-label XOR program then gives $C_{\mathrm{out}} C_{\mathrm{in}}^{-1} = C_{\mathrm{full}}$ on every routed role, including arbitrary scratch roles. Its inverse role merge removes the scalar role permutation by (A.4a). Off eligibility every phase is identity and the same merge returns identity. No sector is extracted and no clean payload value is used in this induction.

Replace each binary frozen XOR mask in the finite router by the weighted two-bank endpoint mask. A selected prefix of u endpoint digits has numeric width at most $u \log R + O(u)$. The reserved dock is borrowed only in a slice wide enough for the current target prefix, and is returned after each elementary update. The whole c_0 dock is never exchanged inside a small child. The one-node toll divided by its current volume V, including exact prefix and tile exception repairs, is bounded by

$$O\Big((\log L)^C \big((e \log R)^\tau + (\log L)^C\big)\Big)$$

for a fixed C. Metadata scans cost $O(M_{\mathrm{rec}}L^C)$ and are absorbed by R_{op}. Parking the other current role streams costs $O(V)$ per child, a fixed additive node toll because W_*, s_* are constants. Descendants never scan parked ancestor streams.

To implement that last assertion, use a fixed set of role tapes, a parking stack and a descriptor stack. Before one child, append only the other current role streams and separators to the parking top, copy the active stream to child I/O, and clear the current work intervals. After the child returns, restore inactive streams by popping right to left and writing right to left into prepared intervals. This preserves their internal order and ends at the saved current-parent boundary. The child's own stack material has already been popped. No operation crosses a parked ancestor. The total passing/return cost per child is $O(V)$ at its parent, and the number s_* of children is a fixed constant. It adds a fixed normalized toll, without multiplying child time. All children keep the full five-bank native coordinate universe and the same low opaque lanes, so polynomial-in-L descriptor work fits in their own volume even when their row count is small. SWAP uses a separate fixed tape family and its own stack; it never calls this phase recursion. The total number of tape families stays finite.

For exact power widths $e = m^k$, with bounded k handled by direct finite tables, the normalized recurrence is therefore

$$F(m^k) \le \frac{s_*}{W_*} F(m^{k-1}) + O\Big((\log L)^C\big((e\log R)^\tau + (\log L)^C\big)\Big).$$

For an arbitrary active count, the supplier partitions the ranks into base-m digit-conditioned consecutive power-width groups and handles those groups in separate passes; their widths form a geometric sum. The group placements are charged separately at full block width. There is no physical gather implicit in a group placement. For its public active-coordinate permutation, decompose each cycle $j \mapsto j+1$ as the two reflections $j \mapsto -j$ and $j \mapsto 1-j$. Each reflection is a disjoint matching of coordinate swaps computable from the whole mask S. On one pair, the chronological trit updates $y \leftarrow y+x$, $x \leftarrow x-y$, $y \leftarrow y+x$, $x \leftarrow -x$ give the swap. Each shear is a disjoint source/target matching from the selector construction of Appendix A, and all masks and quotient labels return at its completed boundary. Thus a placement and its inverse are priced at full h width, in $O(V(h\log R)^\tau \operatorname{polylog} L)$ time. There are only $O(\log h)$ digit-conditioned passes, with a fixed m-dependent constant. These are root placement tolls, rather than costs incurred at each smaller child. There is no ceiling recurrence for arbitrary e. Put $a_0 = s_*/W_* = m^\sigma$ and absorb the fixed logarithmic radix factors in $Q(L) = (\log(L+2))^{C'}$. The exact power recurrence is

$$F_k \le a_0 F_{k-1} + O(Q(L)(m^{k\tau}+1)), \qquad F_0 = O(Q(L)).$$

Since $\sigma < \tau$, its geometric ratio a_0/m^τ is strictly below one, uniformly in L, h and the current row range. Thus $F_k = O(m^{k\tau}Q(L))$. More generally the exponent is $\max\{\sigma,\tau\}$, with an extra factor $k = O(\log L)$ only when they coincide. Summing the separately charged groups and their full-h placements gives $F(h) = O(h^\tau \operatorname{polylog} L)$. This uses the finite phase-H supplier above, not an assumed generic fast H_2 tensor.

Apply the 776-letter word, containing 384 phase letters, to h four-state digits. The permutation part of each monomial is an affine permutation of $\mathbb{F}_2^2$; its tensor power is a constant number of NOT/CNOT mask stages, implemented by the same weighted router. If $M_{\pi(d),d} = m_d$, first scan the source addresses and multiply by $\prod_i m_{d_i}$, then perform that routing. This costs $O(L^{16} + h + R_{\mathrm{op}})$ per address: the already certified public address decoder, $O(h)$ fixed-field products for its coefficient, and a linear scan of the opaque record. Here $h \le L$ and $R_{\mathrm{op}} \ge L^{64}$ in the admissible range, so the scalar stage costs $O(V)$. No stale source-indexed coefficient list is applied at a post-permutation address. Pad the payload to fixed $\mathbb{F}_{16}$ lanes; the complete word has binary entries and returns the auxiliary lanes to zero. Do not crop these lanes inside an A letter, monomial, or intermediate scalar stage.

The completed block applies H_2 to the selected $2h$ endpoint axes and identity to every other native label, endpoint, role, dock, and suffix index. Its ordinary binary transpose is the same completed operator, since H_2 is symmetric. Its inverse is two forward applications, since $H_2^3 = I$.

B.6 Pad lifetimes, block sum, and exponent

The distinct pads have disjoint lifetimes:

- Each private X or W numeric pad is fixed pointwise during all eight updates, excluded from the exception sort, and cropped only after its completed weighted mask including exact repair.
- Each aggregate native-rectangle pad is cropped after the complete rectangle transpose, before entering the role recursion; the inverse aggregate transpose is applied after the block.
- The high role-row pad is retained through the whole fixed H block. Phase calls return its numeric coordinate, and monomial letters leave it unchanged. It is cropped after the binary word is complete.
- The low suffix pad and bundled $\mathbb{F}_{16}$ lanes are retained through the complete block word. The block acts as identity on those suffix indices and has binary entries, so their artificial values return to zero only at that boundary.
- The physical dock is restored after each elementary weighted update. It is not a recursion child field and never accumulates state.

No factors from these pads are reapplied in the recursive child volume. At the outer binary-word boundary, crop the numeric row to its original 2^r values and strip the codec back to R_{op} bits before undoing the r role-bit moves. Then crop the whole spectator suffix and undo the bank-layout exchanges. This order never treats a padded numeric row as an r-bit field, or a padded codec as original index bits.

Partition the t endpoint axes into blocks of at most $2h$; a short final block uses the same supplier at its smaller width, and one unpaired axis is a direct two-row map. Because $h \le t/16$, each block has the required three disjoint selector banks. For sufficiently large L, $L - 2 < \log_2 M \le t \log_2 R$ makes the second term in the definition of h the smaller one. Thus $h = \lfloor L/(40\ell_R) \rfloor \ge L/(80\ell_R)$. Since also $t \le L/(3\log_2 L)$, the number of blocks is $O(\log R) = \mathrm{polylog}(L)$. Their completed operators commute and restore all borrowed indices. Initial/final aggregate layouts cost $O(VL^\gamma)$ per block. Summing block and role costs gives

$$O\Big(VL^{\max\{\sigma,\tau,\gamma\}}(\log L)^{C''}\Big).$$

With $\tau = 1 - 2^{-36}$, $\sigma < \tau$ and $\gamma = 1/(2-\tau) > \tau$, the same construction therefore gives

$$O(VL^\gamma \,\mathrm{polylog}\, L).$$

Retaining this parameter avoids repeated numerical weakenings at later compiler interfaces. More generally their exponent is $\max\{\sigma, \gamma\}$. Bounded remaining parameters use direct finite maps and enlarge the single uniform constant.

B.7 Dependencies and exact scope

The construction depends on:

1. the interchange construction of Appendix D, with its fixed tape and arbitrary-gap interface;
2. exact stable extraction/sorting of public exceptional address keys, with its full-payload merge-sort bound shown above and further implementation details in `supporting/OPAQUE_RECORD_PACKED_ROUTING_AUDIT.md` and `supporting/H2_FINAL_PACKED_ROUTER_CERTIFICATE.md`;
3. the weighted tile identity and safe-set/repair proof above, with its earlier expanded ledger in `MIXED_RADIX_WEIGHTED_TILE_MASK_LEMMA_20261030.txt`;
4. the finite $A = C \oplus [1]$ role/frame supplier and the certified $H_2 \otimes H_2$ word proved in Appendix A.

The Euclidean aggregate-rectangle schedule and its fixed-tape return protocol are given in `supporting/ntt_base_five/fixed_tape_primitive_reconstruction.md` and `supporting/ntt_base_five/chunk_swap_uniform_completion_audit.md`. The weighted tile supplier's control bits must be frozen during its eight updates and disjoint from target, auxiliary, and dock. The canonical bank schedule above provides that disjointness for every individual shear; a selector that reads its changing auxiliary would fall outside the lemma. The supplied role proof must also be read with its exact source correction and pad lifetimes; a generic circuit-size bound does not imply this tape recurrence.

This appendix proves only the canonical adjacent-pair tensor on the stated mixed-radix family using these explicit constructions. It gives no cheap conversion for an arbitrary perfect matching on a native digit. It also does not price an ordinary cyclic DFT-to-prime-grid permutation or the scalar arithmetic required for a full multiplication construction. Those are separate interfaces and must not be inferred from the endpoint theorem.

C Common fixed-field words and exact-width replay

This appendix makes the local characteristic-two substitution explicit. It does not import the complex convolution service from [6].

Newton symmetry over a ring. Let P have as column k the monomial coefficients of $\prod_{a<k}(Z - u^a)$. It is unit upper triangular, and evaluation gives $F_p(u)P = N$. With $D = \mathrm{diag}(N_{kk})$ and $L = ND^{-1}$, this is the unit-lower/diagonal/unit-upper factorization $F_p(u) = LDP^{-1}$. Such a factorization with invertible diagonal is unique over a commutative ring: comparing two makes a unit lower triangular matrix equal an upper triangular one, identifying the diagonal and then the two unit triangular factors. Since $F_p(u)$ is symmetric, its transposed factorization is another of that form. Therefore $P^{-1} = L^T$ and $F_p(u) = ND^{-1}N^T$. The pivot inverses are exactly the units proved in the main text; no field cancellation has been assumed.

A shared eight-H schedule. Write $D_a = \mathrm{diag}(a, 1)$, with $a = \eta$ or η^2. Direct multiplication in $\mathbb{F}_4$ gives

$$H^3 = I, \qquad HD_\eta H^2 D_{\eta^2} H = \mathrm{diag}(\eta^2, \eta)J,$$

and

$$W_4 := HD_\eta HD_{\eta^2} HD_\eta HD_{\eta^2} = I.$$

Thus $I = W_4W_4$, J is its displayed four-H word with a leading unit diagonal followed by W_4, and $H = H^4W_4$ all have eight H letters with unit diagonals in common slots. Identity diagonals fill missing gaps. The formulas include all identity pairs and never select a zero scalar. Coefficients may depend on a pair label because every intermediate letter preserves that label. Controlled coefficients are evaluated only at the completed K_T boundaries described in the main proof.

Formal convolution over a coefficient algebra. For comparison, Cantor–Kaltofen [3] give sharper classical size bounds and logarithmic arithmetic depth. Their sparse integer *stage* matrices compose to a formal bilinear identity; the composed matrices need not be sparse. We use the following elementary triadic construction instead, so its descriptor generation and coefficient types can be checked directly.

Lemma 12 (Elementary formal convolution)**.** *Over any commutative $\mathbb{F}_2$-algebra A, ordinary multiplication of two length-v polynomials has a division-free formal bilinear circuit with size $O(v \log^8(2v))$, depth $O(\log(2v))$, binary fan-in, and deterministic polynomial-v wire-descriptor generation.*

Proof. Use schoolbook products below the fixed threshold $v_0 = 6^6$, with balanced output sums. Otherwise let b be the least power of three with $b^2 \geq v$, put $s = \lceil v/b \rceil$, and let t be the least power of three with $t \geq 2s - 1$. Then

$$\sqrt{v} \leq b < 3\sqrt{v}, \quad s \leq b, \quad t \leq 3b, \quad t < 6s, \quad t \mid 3b.$$

Work formally in $R = A[Y]/(Y^{2b} + Y^b + 1)$, free of rank $2b$ over A. The polynomial is Φ_{3b} reduced modulo two. Its roots over $\overline{\mathbb{F}}_2$ have exact order $3b$: they satisfy $Y^{3b} = 1$, and none satisfies $Y^b = 1$ since the defining polynomial then equals one. Every proper divisor of $3b$ divides b. Therefore $Y^j - 1$ is coprime to this polynomial for $0 < j < 3b$. The corresponding Bezout identity extends from $\mathbb{F}_2$ to A, proving principality of Y and of $\zeta = Y^{3b/t}$ at order t. The inverse DFT uses ζ^{-1}; its normalization is one because t is odd.

Encode the input as $\sum_{i<s} A_i(Y)Z^i$, with $\deg A_i < b$, and similarly for the other operand. Take their t-point DFTs over R. At a frequency the value is *one* R element, represented by $2b$ coefficients from A; no unevaluated Z variable remains. Multiply two such values by one recursive ordinary length-$2b$ product over A, and reduce the degree-$\leq 4b - 2$ result modulo $Y^{2b} + Y^b + 1$. For $0 \leq r \leq 2b - 2$, the high monomial reductions are

$$Y^{2b+r} = \begin{cases} Y^{b+r} + Y^r, & r < b, \\ Y^{r-b}, & b \leq r \leq 2b - 2. \end{cases}$$

They cost $O(b)$ additions in constant depth.

The inverse DFT recovers $C_j(Y) = \sum_{i+k=j} A_i(Y)B_k(Y)$ without wrap in j, since $t \geq 2s - 1$. Each true C_j has degree at most $2b - 2$, below the quotient degree, so reduction has lost no coefficient information. Reconstruct the ordinary product as $\sum_j C_j(X)X^{bj}$: adjacent length-$2b$ blocks overlap, and their overlaps are added, not concatenated. Crop the known padded zero tail to length $2v - 1$.

Multiplication by a root power in R pads a $2b$-coordinate vector to $3b$, rotates, and folds $(a_0, a_1, a_2) \mapsto (a_0 + a_2, a_1 + a_2)$. Each coordinate uses at most two inputs. A radix-three butterfly therefore has $O(b)$ binary additions and constant depth over A. Explicitly, at transform length $t = 3u$ split inputs as x_{3j+r}, $r = 0, 1, 2$. Compute their three length-u transforms $B_r(q)$ with root ζ^3, then print

$$\widehat{x}_{q+\ell u} = \sum_{r=0}^{2} \zeta^{r(q+\ell u)} B_r(q), \qquad 0 \leq q < u, \quad \ell = 0, 1, 2.$$

This recursively specifies every butterfly and twiddle index. The two forward and one inverse transforms, pointwise folds and reconstruction give, for an effective absolute constant C,

$$S(v) \le tS(2b) + Ctb\log(2t), \qquad D(v) \le D(2b) + C\log(2t).$$

These counts are in A gates, not unit-cost R operations. Moreover

$$2bt < 12bs \le 12(v+b) \le 24v, \qquad 2b < 6\sqrt{v} \le v^{2/3} \quad (v \ge v_0).$$

Induction using $S(v) \le Kv(\log_2 v)^8$ above the direct branch contracts the recursive term by

$$24(2/3)^8 = 6144/6561 < 1.$$

The $O(v\log v)$ toll is absorbed by increasing a fixed K; the finite direct branch fixes an effective base constant. This proves the size bound. The logarithms of child lengths contract by $2/3$, so the depth recurrence sums to $O(\log(2v))$.

All nonrecursive maps are the printed linear maps; recursive leaves multiply a left-operand form by a right-operand form. This proves a formal bilinear identity over A, without $x^2 = x$. For generation, compute b, s, t by integer loops, emit radix-three butterflies with rotation/fold source indices, and recursively emit the t children. Each descriptor has $O(\log v)$ address bits; source indices can be obtained by schoolbook arithmetic and sequential list scans in polynomial time. This finite recursive printer is uniform, with no root table, factorization, or coefficient-dependent branch. □

Here is a second explicit way to see that specialization leaves only linear gates. After fixing the kernel, attach to each gate its constant value c_g and its homogeneous degree-one part ℓ_g in the remaining input indeterminates. An input x_i has pair $(0, x_i)$, a fixed scalar a has $(a, 0)$, and the rules are

$$(c,\ell) + (d,m) = (c+d, \ell+m), \qquad (c,\ell)(d,m) = (cd, c\,m + d\,\ell).$$

The constants are prepared once. Only additions and multiplication by these fixed constants remain in the linear DAG. The rules are formal truncation modulo total degree at least two; they use no Boolean relation $x_i^2 = x_i$. Since fixed-kernel convolution is homogeneous linear, its output is exactly the degree-one part. The transformation changes size and depth by constant factors.

Bounded-degree copy trees, including copies of inputs and outputs, cost size proportional to the number of use edges. Replacing each gate's fan-out by its balanced tree adds at most $O(\log v)$ depth per original circuit layer. The deliberately loose linear-DAG bounds are therefore

$$g(v) = O(v\log^8(2v)), \qquad H(v) = O(\log^2(2v)).$$

All wire and scalar descriptors have polynomial-time generation in v and the explicit coefficient dimension. Kernel-specific constant propagation is charged once in that setup, with the actual coefficient words retained. This construction is independent of the faster multiplier being proposed.

Replay on arbitrary dirty coordinates. Let a linear DAG implement $M : A^e \to A^a$, and assign a distinct borrowed coordinate to each of its g gates. The physical source coordinates x, target coordinates y, and borrowed coordinates z are pairwise disjoint. Every output is a designated wire; any additions needed to form it and any input/output copy trees are included in the counted DAG. Thus the output-read edges also have bounded degree. In the notation below, F, D name these actual port reads, rather than uncosted dense matrix

applications. Replace gate assignment by addition of its predecessor form into that coordinate, in topological order. For initial borrowed vector z this sweep has form $z \mapsto Uz + Tx$, where U is unit lower triangular. If the prescribed reads are $Fz + Dx$, zero initialization would compute the DAG, so $FT + D = M$. Sweep, add these reads to y, and undo the sweep. The net increment is $Mx + FUz$, while z is restored. Now sweep with every direct edge from x omitted, including edges to input-copy-tree roots, and add only the borrowed-coordinate read FUz to y, and undo again. In characteristic two the two FUz terms cancel. This realizes $(x, y, z) \mapsto (x, y + Mx, z)$ for every initial z, including coordinates correlated with current sources and targets.

At each DAG depth destinations are fresh gate coordinates and predecessors are earlier ones. Bounded fan-in and fan-out give a bounded-degree graph of shear edges. A constant edge coloring makes each color disjoint pairs. Apply its colors before advancing to the next depth; undo in reverse depth and color order. The output-port edges into the distinct y coordinates have the same constant-color bound. In the second read omit every direct x-to-output edge as well. Zero-coefficient edges may be omitted throughout. The four sweeps and two output rounds thus use $O(H + 1)$ shear layers, on exactly $e + a + g$ coordinates. No clean gate values or fresh zero coordinates are assumed. Other parent coordinates can be selected as borrowed gates, even if they contain data from an earlier chunk operation; the replay restores them exactly. A source/target alias is outside this contract.

Explicit rectangular displacement reconstruction.

Lemma 13 (Exact-width unit Toeplitz word)**.** *Over any commutative $\mathbb{F}_2$-algebra A, a unit lower triangular Toeplitz matrix T_v has a word on exactly its original v coordinates of $O(\log^{19}(2v))$ layers. Each layer consists of shears $(x, y) \mapsto (x, y + \mu x)$ on disjoint ordered pairs. No clean coordinate is required. Descriptors and coefficients can be prepared by polynomially many operations on indices, finite lists, and public coefficient-algebra elements.*

Proof. Write $(T_v)_{ij} = h_{i-j}$ for $i \geq j$, with $h_0 = 1$. Its reciprocal sequence is prepared by $g_0 = 1$ and $g_k = -\sum_{j=1}^{k} h_j g_{k-j}$; no coefficient inversion is needed. For $v \geq 2$ split at $s = \lfloor v/2 \rfloor$ and put $r = v - s$. The matrix has blocks

$$T_v = \begin{pmatrix} T_s & 0 \\ B & T_r \end{pmatrix}.$$

After applying the two diagonal children in place, a cross update $E = BT_s^{-1}$ gives this matrix exactly. Its entries are

$$E_{ij} = \sum_{\nu=j}^{s-1} h_{i-\nu} g_{\nu-j}.$$

For consecutive row/column chunks, the interior difference is

$$E_{ij} - E_{i-1,j-1} = -h_{i-s} g_{s-j}.$$

Let M be that rectangular chunk and let J_a, J_e be lower shifts. Its difference $\mathcal{D} = M - J_a M J_e^T$ is the displayed outer product plus first-row and first-column corrections, hence the sum of three explicit outer products. More explicitly, let the chunk's absolute row and column indices start at $i_0 \geq s$ and $j_0 < s$, and use zero-based local indices. Set

$$v_i = -h_{i_0+i-s}, \qquad w_j = g_{s-j_0-j}, \qquad r_j = M_{0j} - v_0 w_j,$$

and set $c_0 = 0$, $c_i = M_{i0} - v_i w_0$ for $i > 0$. Then, with e_0 the first standard vector of the appropriate length,

$$\mathcal{D} = vw^T + e_0 r^T + c e_0^T.$$

The interior follows from the displayed difference identity, the second term corrects the complete first row, and the third corrects the remaining first column without counting $(0,0)$ twice. Every entry is given by finite sums and products of the prepared Toeplitz and reciprocal coefficients. This uses no field rank test. Nilpotence gives $M = \sum_{l\ge 0} J_a^l \mathcal{D} (J_e^T)^l$. For one term vw^T, its action on x is

$$b_l = \sum_{j=l}^{e-1} w_{j-l} x_j, \qquad c_i = \sum_{l=0}^{e-1} v_{i-l} b_l.$$

Here $v_k = 0$ outside $0 \le k < a$. For literal convolution windows put $W(Z) = \sum_{j<e} w_j Z^j$, $X^{\mathrm{rev}}(Z) = \sum_{j<e} x_j Z^{e-1-j}$ and $V(Z) = \sum_{i<a} v_i Z^i$. The polynomial with coefficients b_l is

$$\mathrm{rev}_e\big((W X^{\mathrm{rev}}) \bmod Z^e\big),$$

and the output polynomial is its product with V, reduced modulo Z^a. Here $\mathrm{rev}_e P = Z^{e-1} P(Z^{-1})$ at fixed padded width. The two ordinary products have output lengths at most $2e-1$ and $a+e-1$, respectively. If $a, e \le b$, both are at most $2b-1$. Their padding zeros are formal constants; they introduce no extra runtime source coordinates. Reversal and selection specify DAG ports. The three displacement terms therefore use six fixed-kernel convolution circuits and their sum, with total gate bound $O(b \log^8(2b))$ and depth $O(\log^2(2b))$ including all intermediate and copy coordinates.

At setup enumerate $b = 1, \dots, v$, partition both sides of the cross block into consecutive chunks of width at most b, and print the actual DAG for every pair, including ragged final chunks and its input/output copies. Choose the largest b for which $e + a + g \le v$ for every pair. For a uniform fixed C the count is $e+a+g \le 2b + Cb\log^8(2b)$. Thus $b_0 = \lfloor cv/\log^8(2v) \rfloor$ passes for sufficiently large v and a sufficiently small fixed $c > 0$; the chosen size is at least b_0. If none passes, use the direct triangular shear word, updating targets in descending order from their earlier source coordinates. This branch occurs only in a fixed finite range.

For each accepted pair its source chunk lies in $[0, s)$ and its target chunk in $[s, v)$, hence they are disjoint. Choose the g borrowed coordinates from their complement inside this same v-coordinate parent. Replay restores that complement before the next pair. The source half is unchanged by every completed cross update, so the sum of all chunk updates is exactly E. There are $O(\log^{16} v)$ chunk pairs, each with $O(\log^2 v)$ replay layers. The diagonal children use only their own disjoint intervals. Run their matching layers in parallel, padding the shorter word with identity layers. Their union is again a matching. Thus

$$d(v) \le \max(d(\lfloor v/2 \rfloor), d(\lceil v/2 \rceil)) + O(\log^{18}(2v)), \qquad d(v) = O(\log^{19}(2v)).$$

All coordinates belong to the original v-coordinate parent. Preparing and counting graphs, reciprocal coefficients, outer-product generators and edge colors uses only polynomial-size finite lists and polynomially many public coefficient operations. The search is setup for the public kernel, never a per-payload-record computation. □

The Newton Toeplitz sequence is $h_j = H_j^{-1}$, so $h_0 = 1$ as required by Lemma 13. Transposing a layer word reverses its layer order and each shear's source/target direction; this supplies the upper triangular word on the same coordinates at the same layer count. This transposition is on A-coordinate indices and retains each coefficient μ. It is a descriptor operation, not a physical array transpose. Consequently $F_p = N D^{-1} N^T$ uses two such Toeplitz words and a fixed number of unit-diagonal stages. Synchronize different prime axes by identity matching layers. With $p \le R_0$, preparing all their words and actual B-coefficient tables has bit cost polynomial in R_0, D', L_0, as in the coefficient setup below. Each matching layer is then replaced by the fixed local slot word in the main text; these abstract layer counts do not assign unit bit cost to growing scalars or tape movement.

The actual constant extension. Every irreducible factor of $\Phi_{\alpha'}$ has degree $r = \mathrm{ord}_{\alpha'}(2)$, so S is a product of copies of $\mathbb{F}_{2^r}$. A factor tensored with $\mathbb{F}_4$ splits into $\gcd(r, 2)$ fields of degree $r_B = \mathrm{lcm}(r, 2)$. Therefore the same Frobenius exponent in the main proof selects $\mu = 1$ in every field component of B, with $r_B \le 2D'$. Computing r by repeated multiplication by 2 modulo α' needs at most D' iterations and has polynomial small setup cost. Exponentiation for the selector needs $O(r_B)$ ring products. The fixed $\mathbb{F}_{16}$ codec inside the binary H supplier need not embed into B: the completed H acts identically on each of the $2D'$ binary lanes and is thus the required B-symbol map.

The central $\mathbb{F}_4$ in B and a fixed field codec used inside a binary routing/H kernel need not have the same intermediate bit encoding. The replacement is made only at a completed binary kernel: its target matrix is H, I or J on whole symbols and acts identically on every binary lane. It therefore implements the required B-linear map in the caller's original representation. Outer multiplications by elements of B are inserted only after that boundary and always use the separately charged coefficient-ring arithmetic. Even a central $\mathbb{F}_4$ action on two concatenated S lanes can be implemented by a constant number of aligned word scans; no growing basis conversion or free bit transpose is assumed.

Actual growing-word arithmetic and unit setup.

Lemma 14 (Fixed monic remainder at any input width)**.** *Fix a monic binary polynomial f of degree $d \ge 1$ and a public input width $r \ge d$. After polynomial-r setup, the remainder modulo f of any binary polynomial of degree $< r$ costs $O(M_{\mathrm{old}}(r))$ bit steps on the fixed tape model, including all scans and returns.*

Proof. If $r = d$, the input is already reduced. Otherwise set $k = r - d > 0$. The reversal $f^* = X^d f(X^{-1})$ has constant coefficient one by monicity. Prepare $h = (f^*)^{-1} \bmod X^k$ once, for example by its coefficient recurrence with elementary polynomial-time tape arithmetic. For the dividend w put

$$w^* = X^{r-1} w(X^{-1}), \qquad q^* = (w^* h) \bmod X^k, \qquad q = X^{k-1} q^*(X^{-1}), \qquad R = w - qf.$$

All reversals have the displayed fixed padded widths. To prove the formula, write the unique monic division $w = Qf + R_0$ with $\deg Q < k$ and $\deg R_0 < d$. Reversal gives $w^* = Q^* f^* + X^{r-1} R_0(X^{-1})$; the second term is divisible by X^k. Multiplying by h modulo X^k therefore recovers Q^*, so $q = Q$ and $\deg R < d$.

The first product may truncate w^* to its first k coefficients. It has two inputs of length k; the other product has input lengths k and $d + 1$, each at most r. Thus two calls to the independent word supplier cost $O(M_{\mathrm{old}}(r))$, and all reversals, cropping, XOR and head returns add $O(r)$. There is no per-dividend Euclidean algorithm. The prepared inverse length is k, even when $d \ll r$. □

Put $d = D'$ and $f = \Phi_{\alpha'}$. Store a B symbol as the two consecutive d-bit S words in $a + b\eta$. If $P = ac$, $Q = be$, and $T = (a + b)(c + e)$ in S, then

$$(a + b\eta)(c + e\eta) = (P + Q) + (T + P)\eta.$$

Three binary polynomial products modulo fixed f, supplied by Lemma 7, with linear XOR scans, therefore suffice. Their degree-$\le 2d - 2$ outputs use Lemma 14 at input width $2d - 1$. Its reciprocal is prepared at length $d - 1$ for $d \ge 2$; $d = 1$ is direct. This proves $O(M_{\mathrm{old}}(d))$ application time with no dense d-by-d matrix operation. The runtime word supplier is classical integer SS with carry-free binary packing; it is separate from the formal characteristic-two convolution circuit used to construct the local linear DAG above.

The involution $\eta \mapsto \eta + 1$ has norm $N(a + b\eta) = a^2 + ab + b^2 \in S$. For a unit $x = a + b\eta$, its norm is an S unit: it is a product of units in B, and an inverse in the free extension $B = S \oplus S\eta$ supplies an inverse in S by taking its constant coefficient. Thus

$$x^{-1} = ((a + b) + b\eta)\, N(x)^{-1}.$$

Polynomial extended Euclid with f suffices for the norm inverse, without factoring B. Such inverses are prepared once in polynomial time in the small dimensions. The selector exponent $2^{r_B} - 1$ also needs only $O(r_B)$ binary squaring/product steps, with $r_B \le 2d$. For example, from $z = \mu + 1$ repeat $z \leftarrow z^2(\mu + 1)$ exactly $r_B - 1$ times. No possibly nonunit $\mu + 1$ is inverted. The one-time polynomial setup is separate from every runtime scalar application.

Scalar lists on fixed tapes. At a synchronized local slot, each axis has a public table $a_i(x_i) \in B^*$ in that slot's current coordinate order; the tensor diagonal is $\prod_i a_i(x_i)$. Pairing and coefficient choices are axis-local. Jointly controlled CRT coefficients use their different generator and are not covered by this factorization.

Let $M_j = \prod_{i \le j} n_i$, $M_0 = 1$, with axes in physical lexicographic order. Start with the one-word list [1]. For every partial product c on the current list, scan the entire next local table and print $ca_j(0), \ldots, ca_j(n_j - 1)$. Rewind that local table, then advance to the next c. Its rewind costs the size of the child block just printed. Alternate two expansion tapes and erase the previous list at the end of each level. This keeps the produced list in the current native order and uses a fixed tape set.

Level j uses M_j growing-ring products and $O(M_j(d + L))$ copying, counters and returns. Here $2d \ge L$, and all $n_i \ge 2$, so

$$\sum_j M_j < 2M, \qquad \text{cost(generate and apply one tensor diagonal)} = O(MM_{\text{old}}(d)).$$

Apply the final list to the payload stream in lockstep and erase it. Prepare or select each local table once per axis/slot in the polynomial small setup, using one shared archive and one reusable active-table tape; there is no tape per axis or indexed archive lookup per payload record. Rebuild the list from the next slot's tables after completed relabeling, H and J calls. Transporting an old scalar list and presuming alignment is not part of the program.

D Equal-chunk interchange from the finite rational-frame motif

The finite rational-frame network and array-interchange strategy are adapted from OpenAI's Sections 3–4 [5]. The explicit reconstruction below supplies the parameters and tape accounting used in this paper.

This appendix constructs, rather than assumes, a fixed-tape interchange algorithm. For every binary width $u \ge 1$, positive integers P, G, B, and arbitrary B-bit record payload, the machine takes an array in lexicographic order on

$$[P] \times [2^u] \times [G] \times [2^u] \times [B]$$

to the array whose two u-bit fields are exchanged, retaining P, G, B and the internal order of each record. Its time is

$$O(Vu^\tau),$$

where $V = PGB2^{2u}$ is the entire stored volume and one fixed exponent $\tau = 1 - 2^{-36} < 1$ suffices. All counters, fixed-table preparation, copying, row padding, recursive returns, and cleanup are included. No unverified global multiplication algorithm or equal-chunk interchange oracle is a premise.

D.1 The finite scalar routing network

Let $\mathcal{T} = \binom{[g]}{3}$ with $g = 48$. Two triples are adjacent when their intersection has size one. Put

$$v = \binom{48}{3} = 17296, \quad z = 3\binom{45}{2} = 2970, \quad N = v^3, \quad I = 3v^2.$$

For $S, T \in \mathcal{T}$, define the binary maps

$$\begin{aligned} V_{(S,T),T'} &= [T = T'] \, [|S \cap T| = 1], \\ J_{S,(S,T)} &= [|S \cap T| = 1], \\ G_{i,T} &= [i \in T], \qquad R_{S,i} = [i \in S]. \end{aligned}$$

Then $(JV)_{S,T}$ is one exactly when $|S \cap T| = 1$, and $(RG)_{S,T} = |S \cap T| \pmod 2$. If $S = T$, the intersection size is three and only RG contributes one. For distinct triples with intersection one both terms contribute and cancel in $\mathbb{F}_2$; for intersection zero or two both are zero. Hence

$$RG + JV = I_v.$$

One invocation has source x, target y, arbitrary side scratch a, and arbitrary central scratch c. The chronological eight rows are

$$\begin{aligned} &y \leftarrow y - Ja, \quad y \leftarrow y - Rc, \quad a \leftarrow a + Vx, \quad c \leftarrow c + Gx, \\ &y \leftarrow y + Rc, \quad y \leftarrow y + Ja, \quad c \leftarrow c - Gx, \quad a \leftarrow a - Vx. \end{aligned}$$

For initial scratch (a_0, c_0) the target increment is

$$-Ja_0 - Rc_0 + R(c_0 + Gx) + J(a_0 + Vx) = (RG + JV)x = x,$$

and the final scratch is exactly (a_0, c_0). The signs are harmless over $\mathbb{F}_2$, but writing them makes the restoration identity visible. Thus the invocation adds its source to its target for every initial scratch value.

Use three stages. The first is forward from bank X to bank Y, the second is the inverse schedule from Y to X, and the third is forward from X to Y. On arbitrary bank contents,

$$(x, y) \longmapsto (x, y + x) \longmapsto (-y, y + x) \longmapsto (-y, x).$$

Over $\mathbb{F}_2$ this exchanges X, Y. Each triple coordinate participates in each stage once; invocations have disjoint scratch roles. All scratch roles are restored. The resulting finite role permutation ρ exchanges the N data roles and fixes every scratch role. It is the scalar-routing network used below. The finite incidence identity and scratch schedule are also recorded in `supporting/SCALAR_MOTIF_SYMBOLIC_AUDIT.md` and `finite_phase_appendix.tex`.

D.2 Rational frames and exact child count

Address arithmetic is over $\mathbb{Q}$ first and then over $\mathbb{Z}/(q^b)$ for a fixed odd prime q. It is distinct from the payload network over $\mathbb{F}_2$. On $F = \mathbb{Q}^{48}$ use the nondegenerate symmetric form with Gram matrix

$$B_0 = I - J/9.$$

The all-one line has eigenvalue $1 - 48/9 = -13/3$, and its orthogonal complement has eigenvalue one, so the form is nondegenerate. For $U \in \mathcal{T}$, let $t_U = \mathbf{1}_U$. Then

$$\langle t_U, t_V \rangle = |U \cap V| - 1, \qquad \langle t_U, t_U \rangle = 2.$$

Thus P_U, the orthogonal projection onto $\langle t_U \rangle$, is a well-defined rational rank-one projection. Adjacent triples give orthogonal lines. The common-label table for the finite motif follows from these orthogonal direct-sum decompositions; a derivation is in `supporting/ntt_base_five/finite_motif_rank_closed_form_audit.md`.

At one frame stage, let A be the preceding tensor label, P its prefix line, $B = P^{\perp}$, and $L_U = \langle t_U \rangle$. The next center label satisfies

$$(P \otimes L_U)^{\perp} = (B \otimes F) \oplus^{\perp} (P \otimes L_U^{\perp}).$$

Side inclusions use $L_U \perp L_V$ for adjacent U, V. In the fixed physical gate order the common subspace labels are

0	$B \otimes L_U$
1	$B \otimes F$
2	$(B \otimes F) \oplus^{\perp} (P \otimes L_U)$
3	$A \otimes F$
4	$B \otimes F$
5	$(B \otimes F) \oplus^{\perp} (P \otimes L_U^{\perp})$
6	$A \otimes F$
7	$A \otimes F$.

Every label is tensored with the fixed suffix tensor line of the invocation. A side wire named by its physical Y/X neighbor pair (S, T) meets rows $0(S), 2(T), 5(S), 7(T)$; its middle inclusion is $L_T \subset L_S^{\perp}$. A center wire meets rows $1, 3, 4, 6$. Data X meets $2, 3, 6, 7$ and data Y meets $0, 1, 4, 5$. The second stage uses the reversed schedule with logical banks exchanged, so its ordered side-pair names are reversed to retain this physical convention. Scratch input frames are zero, scratch output frames are I_m, and the final data X/Y frames are I_m and $I_m - P_U$. These finite inclusions and wire incidences use no phase-kernel, weighted-mask, H2, or interchange algorithm as a premise.

In the fixed eight-role order the label dimensions are

$$a-1, \quad 48(a-1), \quad 48(a-1)+1, \quad 48a, \quad 48(a-1), \quad 48a-1, \quad 48a, \quad 48a,$$

where $a = \dim A$. The only decreases are the central transitions $48a \to 48(a-1)$, each losing 48 dimensions. With the fixed counts

$$W = 2N + I(vz + 48), \qquad m = 48^3 = 110592, \qquad L = I \cdot 48^2,$$

the signed endpoint dimension telescope and twice the downward variation give the sum of comparable-edge ranks $Wm - 2N + 2L$.

Complete physical edge ledger. The inverse middle stage runs logical rows $7, 6, \ldots, 0$ with source Y and target X. Its physical gate incidences are nevertheless Y-side, Y-center, X-side, X-center, Y-center, Y-side, X-center, X-side, the same chronological order as the other stages. Thus the common-label table is used in its printed order, with the side-pair names reversed; one must not reverse the table along with the logical row numbers. Put $d = 48$ and $a = d^{j-1}$ at stage j. Suppressing the suffix line, the complete scratch dimension paths, including their endpoints, are

physical wire	dimension path	downward loss
side	$0,\ a-1,\ d(a-1)+1,\ da-1,\ da,\ m$	0
center	$0,\ d(a-1),\ da,\ d(a-1),\ da,\ m$	d.

These are geometric inclusions: the middle side edge uses $L_T \subset L_S^{\perp}$, whose nondegenerate residual has dimension $d-2$; the center's downward residual is $P \otimes F$. At an interstage

boundary X enters as $A \otimes L_T$, contained in $(B \otimes F) \oplus (P \otimes L_T)$, while Y enters as $B \otimes L_S$, exactly its next first label. The physical X path is therefore nondecreasing from a formal positive input line to dimension m, and Y is nondecreasing from zero to dimension $m-1$. Their ordinary edge-rank sums are both $m-1$. Each side contributes m and each center $m+2d$. Since there are $I \cdot 48$ centers, summing gives directly

$$2N(m-1) + (W-2N)m + 2I \cdot 48d = Wm - 2N + 2L.$$

The actual negative X source, treated next, replaces its formal zero-rank first edge by one rank-one edge; it does not alter the ordinary interstage inclusions.

At each of the N exceptional X sources choose input frame $-P_U$; the first gate frame is $+P_U$. Their difference is $2P_U$, of rational rank one. Since the address modulus will have odd characteristic, this is not zero. These N source corrections add N recursive children, so the full child count is

$$s = Wm - N + 2L.$$

For every routed role w, the endpoint labels obey

$$M_{\text{out}}(\rho(w)) - M_{\text{in}}(w) = I_m.$$

For an X data role the endpoint difference is $P_{U^\perp} - (-P_U) = I_m$; the other data and scratch endpoint pairs are the corresponding zero/full or complementary-label pairs in the finite common-label table. For comparable labels, the rank of the projection difference is the dimension change. The only equal-dimension exception is the N first X edges, already counted as $2P_U$ of rank one. Thus no source correction is hidden by the endpoint identity. At each role edge with label difference A, implement the corresponding address-frame change. The scalar routing sends role w to $\rho(w)$; the displayed endpoint identity then leaves the same target shear on data and scratch roles alike.

The exact fixed counts reduce to

$$\frac{W}{N} = \frac{9633881}{1081}, \qquad \frac{L}{N} = \frac{432}{1081}, \qquad 1 - \frac{s}{Wm} = \frac{217}{9633881 \cdot 110592}.$$

The integer inequality

$$217 \cdot 2^{36} > 12 \cdot 9633881 \cdot 110592$$

and $m < 2^{17}$ give $\ln m < 12$ and $1 - s/(Wm) > 12 \cdot 2^{-36}$. From $e^{-x} > 1 - x$ it follows that

$$s/W < m^\tau, \qquad \tau = 1 - 2^{-36}.$$

All constants here are fixed. These are exact rational counts, not a finite-size matrix-rank inference.

D.3 Choosing one odd address prime

There are finitely many rational label matrices, projections, and frame differences. For each edge matrix the construction below gives a fixed factorization into lower-triangular matrices and a partial permutation. Collect the denominators of all entries and the numerators and denominators of every nonzero diagonal unit in these factorizations and their inverses. This is a finite set of nonzero integers. Choose one odd prime q dividing none of them. Such a prime exists by Euclid's argument; trial division through primes finds one after finitely many steps. This is fixed finite setup, independent of b, u, P, G, B. The prime and the rational matrices may equivalently be constants of the finite machine.

All rational matrices now reduce to $\mathbb{Z}/(q^b)$ for every b. The chosen triangular diagonal entries remain units. Each fixed factorization remains an exact identity, with the same a partial permutation pivots. No data-dependent division or prime oracle is used. Because q is odd, the exceptional source matrix $2P_U$ still has its one nonzero pivot and costs one child swap.

D.4 Fixed-radix tape operations

The following operations use fixed q and fixed tape families.

Counters and records. Use little-endian ripple counters with the head returning to the low end. Over a run of A increments, or a run of A decrements, of an r-bit counter from an arbitrary initial state, bit j is visited $O(1 + A/2^j)$ times, so the total cost is $O(A + r)$. The width term is necessary: one increment from $2^r - 1$ can visit all r bits. For a nonempty range of n values, use its own $r = \max\{1, \lceil \log_2(n+1) \rceil\}$ active counter width and initialize/reset it at the prescribed endpoint. A full scan, initialization and return then cost $O(n + r) = O(n)$. To test termination without a full-width comparison after each step, initialize a countdown to $n - 1$, execute the loop body, then decrement and stop precisely on borrow out of its marked r-bit interval. The last decrement from zero performs the end test while visiting those bits. This gives exactly n bodies, including $n = 1$. A needed forward index uses a separate amortized counter. Range bounds are prepared once in their own canonical widths; each reset copies only that bound, not a full ambient-width register. Stale cells outside this active interval are not rescanned at every reset. With a fixed number of nested ranges, charge each inner reset to its ensuing nonempty scan. Copying a B-bit record thus costs $O(B)$ including its counter initialization and returns, even for $B = 1$. Recursion saves counters/descriptors at the current stack top; it does not cross an ancestor to fetch a length. Ordinary schoolbook shape arithmetic has a fixed polynomial cost in descriptor bitlength; the retained-range bound below pays it at every recursive call.

More explicitly, traversing a fixed-depth product of nonempty ranges $n_1, \ldots, n_d$, including a final bit-record range if necessary, has counter cost bounded by

$$O\left(\sum_{i=1}^{d}\Big(\prod_{j<i} n_j\Big)(n_i + \log(2n_i))\right) = O\left(d \prod_{i=1}^{d} n_i\right).$$

For the current fixed matrix dimension, d is fixed, even when prefix, gap and suffix ranges are arbitrarily large. All array bits remain raw payload: counters and boundary markers are on work tapes, with no growing header added to each record. Rewinding the fixed number of current split tracks costs their combined volume and stops at their current stack boundaries.

Digit moves and pointwise maps. Split a stream into q tracks by one selected radix-q digit, preserving order on each track, and merge in the desired digit order. A single digit move costs $O(V)$. For a fixed permutation of two digit positions, split into q^2 tracks and merge according to the two preimage digits; this also costs $O(V)$. A fixed pointwise finite-alphabet map on aligned tracks costs $O(V)$. The number q^2 is large but fixed, not input dependent.

Rotations controlled by earlier fields. Let the target field have $Q = q^b$ values and follow every field used to compute its shift. Within each fixed earlier-address fiber, it consists of Q consecutive blocks of suffix records. For shift $a \in [Q]$, copy the first $Q - a$ blocks to one stream and the last a to another, then write the latter before the former. This realizes $y \mapsto y + a \pmod{Q}$ in $O(V)$ record motion. Computing the offset by schoolbook arithmetic on the fixed number of b-digit controls costs $\mathrm{poly}(b)$ per fiber. Each fiber contains q^b blocks, so this metadata is absorbed into its own volume. Arbitrary spectators and payload remain inside each block. The enumeration retains separate counters for the selected controls. Large spectator counters are advanced or reset by the preceding product-scan rule; their full values are not decoded or copied into the offset calculation. Thus the bound is $O(V)$ even for $b = 1$,

with arbitrarily large prefix, intervening gap, or suffix. This argument relies on reading only the fixed number of selected b-digit fields. It does not authorize an offset depending on the entire ambient address at the same cost.

Fixed rational units. For a fixed positive integer c coprime to q, partition $y \in [Q]$ into its c monotone pieces indexed by $j = \lfloor cy/Q \rfloor$. On a piece the output is $y' = cy - jQ$ in increasing order. Since Q is invertible modulo c, for each output value y' the unique piece is determined by $j \equiv -y'Q^{-1} \pmod{c}$. Split by these fixed c classes and merge their increasing streams in output order. This is $y \mapsto cy \pmod{Q}$ in $O(V)$, including piece endpoints and counters. Prepare the endpoints and $Q^{-1} \bmod c$ in polynomial-b time; the output selector needs only $y' \bmod c$, a fixed finite-state counter updated once per suffix block. It performs no full b-bit multiplication or global-address decoding at each record. The same residue split on a supplied output stream constructs the inverse permutation directly. Negation modulo Q fixes zero and reverses the other $Q - 1$ blocks; a stack reversal plus reversal within each block costs $O(V)$. A fixed rational unit c/d is the composition of integer multiplication by $|c|$, inverse multiplication by $|d|$, and a possible negation. The prime q avoids c, d.

D.5 Lower-triangular frames and rank-many child swaps

For a fixed rational matrix A of rank a, eliminate its topmost nonzero row using the rightmost nonzero pivot. Clear entries to the left by lower-triangular column operations and entries below by lower-triangular row operations. Remove the isolated pivot row and column and continue in the remaining ordered indices. This yields

$$A = E_1 \Pi E_2,$$

where E_1, E_2 are nonsingular lower-triangular matrices and Π has exactly a ones in distinct rows and columns. Reversing the elimination proves the factorization; nonsingularity of A is not required. Concretely, if the pivot is in row i_0, column j_0, clearing a column $j < j_0$ adds a multiple of column j_0 to column j, a lower-triangular right operation. Clearing a row $i > i_0$ adds a multiple of row i_0 to row i, a lower-triangular left operation. Normalize the pivot by a nonzero diagonal scale. Previously isolated pivot rows and columns are unchanged, since their entries in the remaining submatrix are zero. Retaining the original order on the remaining indices preserves triangularity. Thus $LAR = \Pi$ with invertible lower-triangular L, R, and $E_1 = L^{-1}, E_2 = R^{-1}$. The partial permutation need not be diagonal.

Lemma 15 (Signed factorization over every address power)**.** *For each actual directed rational frame difference $A = M_j - M_i$ of rank a, the chosen fixed prime gives, for every $b \geq 1$,*

$$A_b = E_{1,b} \Pi E_{2,b} \quad \text{over } R_b = \mathbb{Z}/(q^b),$$

with both triangular factors invertible. The image and kernel of A_b are free direct summands of ranks a and $m - a$. The same a unit pivots implement the signed address shear; no field assumption on R_b or invertibility of a data coordinate is required.

Proof. The finite rational identities reduce entrywise since q avoids every entry denominator. Each triangular diagonal remains a unit by the same fixed exclusion, so both factors are invertible over every R_b. Multiplication by those invertible factors identifies the image and kernel with those of Π, whose nonzero entries are literal ones. In particular these are coordinate direct summands of the stated ranks. This is an explicit factorization certificate over a ring, rather than a rank test over a field.

Factor the *signed* matrix A before reduction. Its negative can equally be factored as $(-E_1)\Pi E_2$, absorbing the sign in unit diagonals. For the source correction $2P_U$, the factor

two is likewise absorbed in one triangular factor; q is odd. It retains one pivot and one child. No sign or factor two is reduced using the separate characteristic-two payload arithmetic. □

A lower-triangular coordinate change within a bank is performed in descending coordinate order. At target coordinate i, the controls with index $j < i$ still have their original values and physically precede the target. Apply its fixed diagonal unit, then add the fixed linear combination of those earlier coordinates by later-target rotations. This costs $O(V)$ per coordinate; the dimension m is fixed.

Define, for vectors of m radix-q^b digits,

$$\Phi_A(H, D) = (H + AD, D).$$

Its induced forward record pushforward is

$$(\mathcal{P}_A f)(H, D) = f(H - AD, D), \qquad \mathcal{P}_A \mathcal{P}_B = \mathcal{P}_{A+B}, \qquad \mathcal{P}_A^{-1} = \mathcal{P}_{-A}.$$

Here the coordinate signs are in R_b; pointwise payload addition remains XOR over $\mathbb{F}_2$. To implement it, first change D by E_2, change H by E_1^{-1}, add ΠD to H, undo the H change by E_1, and finally undo the D change by E_2^{-1}. The result is $H + E_1 \Pi E_2 D = H + AD$ with the original D restored. The triangular operations are linear volume. For each one in Π, the cross-bank shear $H_i \leftarrow H_i + D_j$ is implemented in the physical order (earlier field H_i, later field D_j) by

$$D_j \leftarrow D_j + H_i; \qquad \mathrm{swap}(H_i, D_j); \qquad D_j \leftarrow H_i - D_j.$$

On (h, d) this is $(h, d) \mapsto (h + d, d)$. The first and last operations target later fields and cost $O(V)$. The middle operation is one recursive interchange of two b-digit fields across arbitrary gap and suffix. Thus Φ_A costs exactly a recursive child swaps plus $O(V)$ other work. Its inverse is Φ_{-A} and has the same bound. Lemma 15 justifies these same fixed factors for singular, negative and source-correction matrices at every b.

D.6 Exact power widths, role frames, and tape recurrence

First let each chunk have width $e = m^k$ radix-q digits. Keep an independent row field of size R divisible by W^k, before both chunks; arbitrary prefix, gap, suffix, and spectators between the row and first chunk are retained. For $k = 0$ the two q-digit chunks are exchanged by the fixed split/merge construction in $O(V)$.

For $k > 0$, split each chunk into m consecutive width-b fields, $b = e/m$, and denote their vectors by H, D. Split whole rows cyclically into W role streams inside each earlier prefix, stream w taking rows $r = Wa + w$. Each role has the full H, D ranges and all spectators, and row count R/W divisible by W^{k-1}. On each finite scalar-network edge from label M_i to M_j, apply $\Phi_{M_j - M_i}$ to its role stream, then perform the specified pointwise XOR gate on aligned payloads.

For input array f_w on role w, use the logical interpretation $\mathcal{P}_{-M_{\mathrm{in}}(w)} f_w$. This is a proof invariant, not a preprocessing pass. The frame change on an edge is common to its aligned XOR operands and commutes with pointwise addition. The scalar network routes role w to $\rho(w)$; its physical output there is

$$\mathcal{P}_{M_{\mathrm{out}}(\rho(w)) - M_{\mathrm{in}}(w)} f_w = \mathcal{P}_I f_w.$$

Equivalently, with block-diagonal input/output pushforward frames $\mathcal{D}_{\mathrm{in}}, \mathcal{D}_{\mathrm{out}}$ and scalar role permutation P, the entire framed word is

$$\mathcal{D}_{\mathrm{out}} P \mathcal{D}_{\mathrm{in}}^{-1} = P(I_W \otimes \mathcal{P}_I).$$

The endpoint difference is I_m on every routed role, including dirty side and center roles. A common reindexing commutes with their XOR gates; no payload value is assumed zero. Merge output role $\rho(w)$ back into input row class w. This undoes the role exchange and applies $H \leftarrow H + D$ to every row, including arbitrary scratch rows, while preserving all spectators.

The finite role network supplies only this earlier-target shear. Complete the chunk interchange with the outer later-target operations

$$D \leftarrow D - H, \qquad H \leftarrow H + D, \qquad D \leftarrow H - D.$$

On input (h, d) the result is (d, h). Only the middle operation uses the framed role network. The first and last operations are later-target rotations and fixed unit maps, each costing $O(V)$. Thus the number of recursive children remains exactly s; it is not multiplied by three.

The finite edge matrices have ranks summing to s. A pivot child is a direct call to the exact power-width entry. More explicitly, its parent role shape is

$$[P][R/W][A][H_1]\cdots[H_m]$$
$$[G][D_1]\cdots[D_m][B].$$

Every formal field has width $b = e/m$, with a separate pretarget spectator range A. A pivot using H_i, D_j has child parameters

$$A_c = Aq^{b(i-1)}, \qquad G_c = q^{b(m-i)}Gq^{b(j-1)}, \qquad B_c = q^{b(m-j)}B.$$

The call is SwapPower$(b, q, P, R/W, A_c, G_c, B_c)$. The row range remains explicit and is divisible by W^{k-1}; all unselected fields remain in the contiguous spectator products. Consequently

$$P(R/W)A_cG_cB_cq^{2b} = P(R/W)AGBq^{2e} = V/W.$$

This power entry introduces no numeric or row pad. It does not call the arbitrary-width or binary wrapper. The pivot factorization therefore gives exactly s recursive calls of volume V/W and width e/m. The finite number of role gates, fixed-rational operations, XOR scans, row splits/merges, and setup work contributes C_1V at the parent. Use separate fixed tape families and a parking stack: before one child, park only the other current role streams at the stack top; pass the active stream to the child; after return, restore siblings by popping and writing right-to-left. Clear only current visited intervals. A descendant never traverses a parked ancestor. Since s, W are fixed constants, all parent parking costs C_sV and does not multiply child time.

With normalized time F_k, the exact recurrence is

$$F_0 \le C_0, \qquad F_k \le (s/W)F_{k-1} + C_1.$$

By the finite count $s/W < m^\tau$, this gives

$$F_k = O(m^{k\tau}) = O(e^\tau).$$

The bound also holds if $s/W \le 1$, since the resulting linear sum in k is at most $O(e^\tau)$. Parent parking uses $O(V)$ time and storage at the current level. At depth j, the child volume is $V_j = V_0/W^j$; the two original full chunk ranges remain present as targets or spectators, so $V_j \ge q^{2e_0}$ for root width e_0. Since $j \le k = O(\log e_0)$,

$$\log V_0 = \log V_j + j \log W = O(\log(2V_j)).$$

Any fixed polynomial of the root descriptor bitlength therefore fits within $O(V_j)$ at the child. Its descriptor lists only the current fields and collapsed spectator intervals, not the chain of ancestor subdivisions.

D.7 Arbitrary q-radix widths without rounding the core

Let the two original chunks have width e radix-q digits. Put $k = \lceil \log_m e \rceil$. Choose the least r_0 such that $q^{2r_0} \geq W^k$. For sufficiently large e, $1 \leq r_0 < e$ and $r_0 = O(\log(2e))$; the bounded smaller range uses direct finite single-digit interchanges and changes only the uniform constant. Write the first chunk as h_+h_- and the second as d_+d_-, with the four lengths specified separately:

$$|h_+| = |d_+| = r_0, \qquad |h_-| = |d_-| = e - r_0.$$

Thus the plus sign always means high digits and the minus sign always means low digits. Starting from

$$[h_+][h_-][G][d_+][d_-],$$

first exchange corresponding high digits individually, obtaining

$$[d_+][h_-][G][h_+][d_-].$$

Then move only the r_0 digits of h_+ left across h_- and G using fixed-radix digit moves, obtaining

$$[d_+][h_+][h_-][G][d_-].$$

This costs $O(Vr_0)$. The joined row field (d_+, h_+) has size $R = q^{2r_0} \geq W^k$. Pad this one independent numeric row range to

$$R' = W^k \left\lceil \frac{R}{W^k} \right\rceil, \qquad R \leq R' < 2R.$$

It is divisible by every W^j needed in the recursion. This private factor below two multiplies the current volume by one constant factor for the whole arbitrary-width call. All children receive that same padded row range and have volume V'/W at the first level (then divide by W per level); the pad is not independently reintroduced at descendant levels. Pad rows are initialized to zero. Completed interchange calls preserve the row coordinate, so these rows remain zero at the crop boundary. Crop the row factor once, after the complete arbitrary-width core, to recover the original volume.

Write $e - r_0$ in base m. At scale m^j there are at most $m - 1$ matching consecutive pieces in each lower chunk. Interchange each corresponding pair by the exact power-width routine, with all other pieces as spectators. The sum of their costs is

$$O(V) \sum_j a_j m^{j\tau} = O(Ve^\tau), \qquad 0 \leq a_j < m.$$

No target piece is rounded up to a larger power. After all pieces, the field order is

$$[d_+][h_+][d_-][G][h_-].$$

Crop the row padding back to R genuine rows before undoing the high digit move. Move h_+ after G by the reverse-direction sequence of r_0 digit moves, obtaining

$$[d_+][d_-][G][h_+][h_-].$$

The two original chunks have been interchanged. The high moves cost $O(V \log(2e)) = O(Ve^\tau)$, and row padding contributes only a fixed factor below two. All spectators retain their order.

For completeness, fixed-radix digit moves use q-way stream splitting and merging; their cost is $O(V)$ each and uses a fixed number of tapes. The $O(\log e)$ piece schedule and shape

arithmetic are polynomial in $\log(2V)$, hence $O(V)$. The full chunk ranges remain in every descendant, which is the volume lower bound used for descriptor costs.

Counters and record movement use ordinary sequential tapes. A little-endian binary counter changes bit j only $O(1 + T/2^j)$ times over T consecutive increments, so a run of T increments, its reset, and return cost $O(T + w)$ for width w. Use a range-specific active width as above; do not restart a full ambient-address counter for each constant-size inner scan. Nested counters and child descriptors are pushed at the current stack top and only that child's interval is popped on return. A B-bit suffix record is copied with an amortized down-counter in $O(B)$, including reset. At each call there are only a fixed number of shape parameters; polynomial work in $\log(2V)$ is $O(V)$ for all positive input volumes after increasing the uniform constant. No counter repeatedly scans an ancestor descriptor.

D.8 Binary fields and final theorem

For binary width u, set $N = 2^u$ and choose the least e with $Q = q^e \geq N$. Then $Q < qN$. Pad each of the two numeric address ranges from N to Q, copying the valid rectangle and filling its complement with zero records. This nested scan costs $O(V')$ where

$$V' = PGBQ^2 < q^2V.$$

Radix-q lexicographic order is numeric order, so no per-record base conversion is used. Apply the exact q-digit interchange above. Its completed coordinate transposition preserves the nested valid rectangle $[N] \times [N]$; hence invalid padding remains zero and can be cropped after the full operation. No crop is performed inside a recursive piece. Because $e = O(u)$ for fixed q, the final time is

$$O(V'u^\tau) = O(Vu^\tau).$$

All operation families have fixed finite tape sets: the q^2 split tracks, the W role streams, reusable I/O/work tapes, and stack/descriptor tapes depend only on fixed constants. Finite rational tables and the avoiding prime are fixed machine data or are generated by terminating constant setup. Counters use ordinary amortized binary increments; record copies and returns cost their full bit volume. Thus this is one uniform fixed-finite-tape construction for all P, G, B, u, including $B = 1$. It proves the local interchange statement without assuming the source preprint's global integer multiplication algorithm. It makes no novelty claim for the finite motif or its routing method.

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